Northern Nut Growers Association Report of the Proceedings at the Fourteenth Annual Meeting

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THE FOURTEENTH ANNUAL MEETING ***

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NORTHERN NUT GROWERS ASSOCIATION (INCORPORATED)

REPORT OF THE PROCEEDINGS AT THE FOURTEENTH ANNUAL MEETING

WASHINGTON, D. C. SEPTEMBER 26, 27 and 28, 1923

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CONSTITUTION

ARTICLE I

Name. This society shall be known as the NORTHERN NUT GROWERS ASSOCIATION.

ARTICLE II

Object. Its object shall be the promotion of interest in nut-bearing plants, their products and their culture.

ARTICLE III

Membership. Membership in the society shall be open to all persons who desire to further nut culture, without reference to place of residence or nationality, subject to the rules and regulations of the committee on membership.

ARTICLE IV

Officers. There shall be a president, a vice-president, a secretary and a treasurer, who shall be elected by ballot at the annual meeting; and an executive committee of six persons, of which the president, the two last retiring presidents, the vice-president, the secretary and the treasurer shall be members. There shall be a state vice-president from each state, dependency, or country represented in the membership of the association, who shall be appointed by the president.

ARTICLE V

Election of Officers. A committee of five members shall be elected at the annual meeting for the purpose of nominating officers for the following year.

ARTICLE VI

Meetings. The place and time of the annual meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the executive committee shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the president and executive committee.

ARTICLE VII

Quorum. Ten members of the association shall constitute a quorum, but must include two of the four elected officers.

ARTICLE VIII

Amendments. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or a copy of the proposed amendment having been mailed by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

ARTICLE I

Committees. The association shall appoint standing committees as follows: On membership, on finance, on programme, on press and publication, on nomenclature, on promising seedlings, on hybrids, and an auditing committee. The committee on membership may make recommendations to the association as to the discipline or expulsion of any member.

ARTICLE II

Fees. Annual members shall pay three dollars annually, or five dollars, including a year's subscription to the American Nut Journal. Contributing members shall pay ten dollars annually, this membership including a year's subscription to the American Nut Journal. Life members shall make one payment of fifty dollars, and shall be exempt from further dues. Honorary members shall be exempt from dues.

ARTICLE III

Membership. All annual memberships shall begin either with the first day of the calendar quarter following the date of joining the Association, or with the first day of the calendar quarter preceding that date as may be arranged between the new member and the Treasurer.

ARTICLE IV

Amendments. By-laws may be amended by a two-thirds vote of members present at any annual meeting.

PROCEEDINGS

AT THE

FOURTEENTH ANNUAL CONVENTION OF THE NORTHERN NUT GROWERS ASSOCIATION

New National Museum, Washington, D. C.

September 26-27-28, 1923.

(In making up this report the transcript of the stenographer's full report has been unsparingly cut, in accordance with the vote of the convention. Copies of the full report are in the possession of the secretary.)

The Convention was called to order at 2 p. m., Sept. 26, 1923, in the New National Museum.

In his opening address the president spoke of the need for increased membership and improved financial condition. He also recommended a return to the old method of combining the secretary and treasurer in one office and that the secretary-treasurer should have a fair salary, suitable quarters, and adequate help. He spoke of his own efforts to increase the usefulness of the association and expressed his fears that they had amounted to very little. He quoted the statement of the editor of the American Nut Journal that what people want to know is whether they can make any money by the cultivation of nut trees. That statement led to a campaign to try to locate in the territory of the association groups of nut trees in profitable bearing. He felt satisfied that there are numerous paying nut orchards, and he recommended a continuance of the campaign for locating such orchards.

The president then went on to instance the experience of Mr. Frederick G. Brown of Salisbury, Mass., at whose place, about two miles from the ocean, there are two Persian walnut trees, 12 to 15 years old, one of them about a foot in diameter and twenty feet high, that have borne for two years. Peach trees will not live at this place. Two miles away at Newburyport is a tree a year or two younger that bore a half peck of nuts last year, and another tree 35 years old in bearing for 15 or 20 years. The nuts were spoken of as of high quality.

He referred to Edward Selkirk of North East, Pa., who has a grove of 250 trees about 22 years old of the Pomeroy variety. Last year the crop was one ton and brought in a little over $500.00. This year the crop is much larger. For best development of the trees the land should be given over entirely to their culture.

The president quoted a letter from E. A. Riehl of Godfrey, Illinois as follows:

My nut plantings are mostly young, many just coming into bearing, while many others have been top-worked to better varieties, so that money returns are not what they would be had I started out planting improved varieties. Part of my aim was to originate better varieties than we had when I began. In this, I think, I have been fairly successful.

My plantings consist mostly of chestnuts. These have sold readily at 35 to 40 cents per pound wholesale. It is rather a hard matter to give any idea as to profit, except that we gathered 23 pounds from one tree five years after topworking on a tree then about three inches in diameter. In 1920, the net return was $1,172.54, in 1921, $1,019.44, in 1922, which was about a half crop, $1,196.81. All this on land so rough no crop could be grown on it but pasture. This year's crop promises to be a full one.

As to walnuts, we have made no record of single trees. The Thomas, by actual test, gives ten pounds of meat to the bushel, which we sold to dealers last season at $1.00 per pound, and could not nearly supply the demand.

Walnut crop here a failure this season. Only a few Thomas trees have a crop.

If the meeting was after nut harvest, I would send the best chestnut exhibit that has ever been shown at any meeting.

H. C. Fletcher of Clarkson, N. Y., was quoted as estimating the nuts produced from two trees each year from 1911 to 1915 as $25 worth. (Presumably these were Persian walnuts, but this was not stated.) In 1916 and 1917 there were about six bushels of nuts, probably $75 worth. In 1918 a market basket full. In 1919 and 1920 about $40 worth, including some trees sold. In 1921 about $50 worth were produced and in 1922 $60 worth of nuts and $30 worth of trees.

In the president's own filbert nursery at Rochester over 300 pounds of fine nuts were produced for which 30 cents a pound were offered by grocerymen.

Mr. W. R. Mattoon of the Forest Service of the U. S. Dept. of Agriculture spoke as follows:

Two years ago, when the Forest Service was planning to get up a bulletin on growing walnut trees for timber, we found the need to include information on the nuts also. Mr. C. A. Reed and I together prepared a manuscript on growing the walnut tree both for timber and for nuts.

It pays to grow walnuts in small groups and singly, rather than in large blocks, for while they have not proven altogether failures when planted in large quantities they have been disappointing. Many of the trees which we planted as close as 6 x 8 feet several years ago, have not given very satisfactory results because they have not had enough light and air. The black walnut grows singly in the forest, although there may be full stands of other trees around it. Our idea is to recommend planting the black walnut in spots around on the farm, in little inaccessible places and on the hillsides, where the soil is good; for the black walnut requires good soil, and we cannot find that quality in large patches, nor is it usual on slopes of ground. So we must put it here and there on the farm, along the fence rows and in various places, but not in groups. The farmer planting in this way becomes its wood which is used in the most expensive furniture. I believe that mahogany is the only other wood so valuable. On the other side of the world they have the mahogany tree for cabinet use, and here in America we have the black walnut, a cabinet wood that is not surpassed.

The present available publications on this subject are limited but we are referring people who inquire about it to Department of Agriculture Bulletin No. 933, "The Black Walnut, Its Growth and Management." That is midway between a technical and a popular bulletin, and it comprises about the only available publication that we have at the present time on the subject of growing the tree. Farmers' Bulletin No. 1123, "Growing and Planting Hardwood Seedlings on the Farm", deals with the black walnut along with other trees. Another publication is Department of Agriculture Bulletin No.

153, "Forest Planting in the Eastern United States," which considers the black walnut along with the other available trees for planting.

MR. OLCOTT: For a small orchard would it be proper to plant 160 to 180 feet apart?

MR. MATTOON: When planted in that way you would get nut production and at the same time, a timber growth. If pruned you get a good log at the base. The small, ten-foot logs from these trees pay as much as you would get for an 18 foot log of a taller tree. For forestry purposes, pruning is a desirable practice.

THE PRESIDENT: But for nut-bearing, what is your opinion?

MR. MATTOON: I should suppose that you would want your orchard trees to be as low-branched as possible, and with as full foliage as possible.

Mr. Bixby (acting as secretary) then read a paper by H. R. Mosnat of Morgan Park, Illinois in which he spoke of the number of doctors interested in nut growing and the need of all men of that character having a hobby of that kind. He thought that the taxes on many farms might be paid out of the profits of nut trees planted on the farms and along the highways. But these nut trees should not be seedling trees. The apple and the black walnut are said to be the only trees that grow in every state of the Union. Nuts were one of the staple foods of our ancestors. We should not be discouraged if we have not yet found the right nut for the East and the Middle West. We should seek them promptly because of the rate at which nut trees are being converted into logs. By next year, he said, he expected to have 25 varieties of black walnuts in his collection including some hybrids. Machines for cracking black walnuts by power are now practically perfect and one firm in that business has cracked about a million pounds in the last few years and expects to treble or quadruple its business this season if supplies can be secured. The trouble with most walnut cracking machines is that they crush instead of crack and small bits of shell are apt to stick to the meats. But there is machinery now to remove these bits of shell. There are wild black walnuts that run 16 to 18 per cent kernels, though the average is only 12-1/2%. It is not always the largest nuts that produce the greatest proportionate weight of kernels. The picking and cracking expense with black walnuts is very little greater than with pecans, but the final cleaning to render the meat absolutely free of shells has been very expensive. Cultivated black walnuts will of course give better results, because they have been selected for easy cracking, have kernels that separate readily from the shell, the product is uniform, and the nuts require much less grading before cracking than the wild black walnuts, where every tree bears nuts differing in size, as in almost every other quality. Figuring 50,000 pounds to the carload it will take about eight carloads of wild black walnuts to make one carload of kernels of the same weight. More and more English walnuts and pecans are being sold in the form of kernels, and black walnuts also will best be sold in kernels. These can be canned in vacuum glass or metal cans, and the housewife will use more nuts when she can get the shell-free meats with her favorite cooking utensil, the can-opener. Confectioners and bakers will take black walnut meats by the carload in preference to other nut meats because they have more flavor, and so "go further."

The growing of black walnuts in a commercial way will require education, but already there is a growing interest. Several of the large weekly publications have, within the last couple of months, carried full page, illustrated articles on black walnuts. One of these, in a magazine of general circulation which is over half a million, within a month resulted in almost one hundred letters asking for additional information, which shows that a great many people want to know more about the possibilities of black walnuts. This interest will certainly increase when profitable black walnut orchards are actually growing and paying good profits. Already men are putting in black walnut orchards or groves of several hundred acres, and one such planting of 1,600 acres is proposed, but it will be partly hardy pecans. This shows rapid development into a real industry of magnitude.

Report of the Secretary.

On March 1, 1923, the treasurer, Mr. W. G. Bixby, handed over to the secretary the funds and books of the association, saying that his time had become so much taken up that he was able to give too little of it to the duties of his office. Thus it became necessary for the secretary to assume the functions of the treasurer as well.

These functions were, in the first place, the payment of the obligations of the association from the funds available. The funds available for current expenses were not sufficient for the payment of these obligations. The secretary therefore took it upon himself to pay these obligations with funds of the association put aside for other purposes. These funds were money received from life membership payments that had been deposited in the Litchfield Savings Society, as a sort of contingent fund, and other funds from the same source held by the treasurer and handed over by him to the secretary. These two funds were completely used up in the payment of current expenses, as will appear in the detailed statement of the secretary.

These funds, however, were still insufficient to pay the current expenses, which were, chiefly, the expenses of the stenographer's report and transcripts of the thirteenth annual convention, at Rochester, and the cost of printing the annual report. The cost of printing the report was paid out of the available funds. The stenographer's bill, amounting to $169.00

originally, but reduced to $135.00 by the stenographer on representation by the officers of the association that the amount was excessive, was paid by Mr. Bixby personally, and the association is indebted to Mr. Bixby in that amount at this moment.

The second function that developed upon the secretary was the management of the membership lists and matters relating thereto, which, though perhaps essentially a duty of the secretary of an association such as this, had been managed by the treasurer since the time when he took over the duties of the secretary in 1918. This had involved quite an expenditure for clerical work. This clerical work would still be an expense to the association, had not one of our members, Mr. H. J. Hilliard, of Sound View, Connecticut, volunteered to do it. Mr. Hilliard was formerly connected with a bank, is entirely familiar with the keeping of accounts, is a man of means and leisure, and I shall take pleasure in offering his name to fill the vacant treasurership. Heretofore, this association has had to pay little or nothing for clerical work which has been done either by the secretary, or by the treasurer and his personal clerical force.

In accordance with the vote of the Rochester convention the secretary drafted two letters, one entitled, "To the State Vice-Presidents of the N. N. G. A. and All Members of the Association"; the other, "To All Women Members of the N. N. G. A. and to All Women Interested, or Interestable, in Nut Culture." Both of these letters were sent to all members of the association, and the letter to women was sent also to a considerable list of women not members. The results of these letters were, so far as the secretary has means of knowing, not over a half dozen letters of appreciation from members, one new woman member, and a letter of appreciation from another woman.

The secretary has reason to believe, however, that the letters were the means of stimulating several of the state vice-presidents to activity in the matter of getting new members, in writing articles for the press and in giving illustrated talks on nut growing. Among those who are known to have given such talks or articles, are Dr. Morris, Mr. Weber, Mr. Spencer, Mr. Smith, Mr. Turk, Mr. O'Connor, Mr. and Mrs. Corsan, Mr. Reed, Mr. Neilson, Wilkinson, Snyder, Matthews, Kains, MacDaniels, Fagan, Kaufman, Rick, Bixby, the secretary, and, doubtless, a number of others.

The secretary has a collection of slides on nut growing which he has lent two or three times to members for illustrating their lectures. It was necessary to provide a box for the safe transportation of these slides which the secretary purchased, at a cost to the association of $8.85. The secretary also furnished a typed, running commentary for these slides and, in one or two instances, has furnished negatives and photographs for making slides and illustrations. The secretary also offers to furnish outlines for lectures or articles, and has a small collection of nuts which is available for lectures.

If the funds were available, it would be possible to enlarge the collections of slides, illustrations and nuts for the use of members who wished to give talks or write articles.

Possibly the suggestion of the secretary was responsible for the formation of a subsidiary association in Rochester. On this a report is desirable from President McGlennon or Mr. Olcott. One or two other members have written of their intention to form subsidiary associations.

A leaflet was also issued by the secretary announcing Mr. Jones' offer to give seedling nut trees as a premium to new members. The demand for these trees not being up to expectation, Mr. Jones very generously sent out five such trees in place of the original offer of one or two. I hope that Mr. Jones will make a report of the number of trees thus distributed. Although the circular distinctly stated that these trees were premiums for new members, many members understood it as an offer for renewal of membership as well, and I think that in every such instance, Mr. Jones himself forgot and sent the trees. A few members, whose names came in too late, were disappointed in not getting trees. Mr. Jones has intimated that it may be possible to correct these omissions this fall. I hope that Mr. Jones will make a statement about this, and I hope also, that the association will not overlook Mr. Jones' liberality in distributing these trees entirely at his own expense.

There have been expressions of regret, and I am sure that many more have felt it, that it has not been possible to go on with the nut contests and the giving of prizes for new and valuable nuts. As there is not likely to be any one else willing to assume the really immense labor involved in the nut contests, conducted as Mr. Bixby has conducted them, I suppose that all we can do is to hope that circumstances will sometime again make it possible for Mr. Bixby to resume these very valuable services for the development of nut culture in the United States. I say intentionally "the United States," because I believe that these services have benefitted the whole country. This fact makes me the bolder in uttering the daring suggestion that perhaps, now that Mr. Bixby has shown the way, and developed exact methods that may be safely followed, which, if I do not misapprehend, is what it states that it desires before presuming to take up any new line of work, the Department of Agriculture itself might consider it a matter worthy of its attention. Professor J. A. Neilson, of the less cautious Canadian Department of Agriculture, is rendering very valuable services of this kind for the Dominion of Canada.

There is evidence that several more state agricultural institutions are giving attention to nut growing. (MacDaniels, at Ithaca; J. C. Christensen, University of Michigan).

There is no need of taking your time now to recapitulate the many things that ought to be done to promote the planting of nut trees and the scientific investigation of nut growing. Dean Watt's address, published in the 12th annual report, and the letter of the secretary to state vice-presidents, contain outlines for these things. The attention of the present convention is more particularly to be given to advocating nut tree planting on a production basis.

Regarding the campaign for new members, perhaps the chairman of the committee on membership will make some remarks. The present membership of the association is 337, if we drop no names this year for non-payment of dues. Of course, those who do not pay their dues should be dropped. But the association has never made any ruling as to how long names should be carried on the rolls. The secretary has been easy in sending copies of the annual reports to members in arrears, hoping that the conscience-stricken recipients would hasten to pay up. But there is no proof that such has been the case, and the secretary would recommend making a rule as to when a member is no longer in good standing, when he should be dropped from the rolls, and what members are entitled to copies of the annual report. The secretary would make the suggestion that there be an amendment to the by-laws to the effect that members who have not paid their dues within three months from the time of their first notification, be sent a second notification to the effect that they are not in good standing on account of non-payment of dues and are not entitled to receive a copy of the annual report; but that all privileges may be restored on payment of dues. At the end of three months from the sending of the second notice, the names of members not in good standing should be dropped. The annual report should be sent only to members in good standing.

Mr. Hilliard asked me what our fiscal year was. I answered that I did not think we had any. It would undoubtedly be a convenience if we are to have a bank man for a treasurer, and a ruling by the association would be in place.

Our accredited list of nut nurserymen is out of date and a new list should be issued. Recommendations as to changes in or additions to that list, should be considered by the members.

It is desirable that the annual reports of the association should be indexed and bound, but no hand has yet been found to do it.

Our ambitions have so far outstripped our sources of revenue that we have come to look on an annual deficit as a normal and defensible thing. I think it is indefensible. I think it is going to have a bad effect on our attendance and our morals if the members have to look forward to what amounts to a good big assessment at every convention. A deficit is not inevitable. The secretary-treasurer was able to report a surplus at the first, fourth, fifth, sixth and seventh meetings. The income from membership dues should be enough to enable the printing of the annual report. But if not I should be in favor of not printing the report until funds were on hand to pay for it.

In rendering an account of the funds of the association I will first state that there is on hand, cash in bank, $84.89. This amount must be charged with the Bowditch hickory prize fund, $25, which leaves $59.89, cash on hand. We owe Mr. Bixby for paying the stenographer's bill, $135.00, and Mr. Olcott for printing, $24.58, a total of $159.58. This makes our deficit $99.69, practically just one hundred dollars.

It should be recalled that in arriving at this result it was necessary to use up our reserve fund from life memberships, amounting to $225.00. If we count that in with the deficit, it amounts to $325.00.

A detailed account of receipts and expenditures is herewith submitted. At the present moment, on account of a rush of other work, on account of difficulties of other kinds, and because of a division of the work between Mr. Hilliard and myself, I am unable to give the exact amount received from memberships and sale of reports and bulletins. This I hope to correct before the annual report goes to press.

RECEIPTS

Turned over by the Treasurer, Mar. 1, 1923:
Money for current expenses $ 89.66
From life memberships 95.00
Bowditch hickory prize 25.00
From Litchfield Savings Society 130.00
Membership dues
Sale of reports and bulletins

EXPENDITURES

Printing report $378.00
Misc. printing and postals 7.50
Clerical hire and postage 47.65
Postage, telegrams, carriage 38.09
Box for lantern slides 8.85

$480.09

Due Mr. Bixby, stenographer's bill $135.00
Due Mr. Olcott, printing 24.00

$159.58

The report of the secretary was adopted.

The following paper was read by the acting secretary as Mr. Neilson was unable to be present:

SOME FURTHER NOTES ON NUT CULTURE IN CANADA.

JAS. A. NEILSON, B. S. A., M. S., Extension Horticulturist, Hort. Expt. Station, Vineland Sta., Ont.

The nut culture activities outlined in the paper presented by the writer at the convention in Rochester were carried on as much as time and means would permit during the past year. The search for nut trees has been continued and has yielded some interesting results. Several valuable trees of kinds already noted have been located and additional species discovered. Among these were five pecan trees which have been growing on the farm of C. R. James at Richmond Hill, a small town fifteen miles north of Toronto. These trees were about fifty years old and appeared to be perfectly hardy, as far as growth was concerned, but owing to the northern location (43.45") seldom produced ripened nuts. The season of 1919, however, was longer and somewhat warmer than most seasons, and a fully ripened crop of nuts was gathered. The nuts are small with a thin shell and a fine sweet kernel. The largest tree in the lot is about 35 feet tall with a trunk diameter of 16" and a spread of branches equal to its height. Another small plantation of pecans was found at Niagara-on-the-Lake on the fruit farm of John Morgan. Some of these trees were of grafted sorts and others were seedlings. Both grafted and seedling trees were making a good growth and appeared to be perfectly healthy.

In as much as the pecan is native to a country having a longer growing season and higher average summer temperatures than southern Ontario, it is quite encouraging to find that these trees will even grow here, to say nothing of bearing nuts. This would seem to indicate that there are possibilities for some of the pecan-bitternut and pecan-shagbark hybrids in southern Ontario where the shagbark and the bitternut grow quite freely.

I also located two excellent shagbark hickories which have fair-sized nuts with thin shell and fine kernels. One of these trees grows about twelve miles west of Simcoe, Ontario, and produces quite a large nut with a shell so thin that it can be easily cracked with the teeth. This particular tree is about seventy feet tall and bore ten bushels of nuts in one season. I have records of several other good hickories and plan to inspect these at the earliest opportunity.

Several more good English walnuts have been located and examined. Among these there is one tree over seventy-five years old which at one time bore thirty bushels of ripe nuts.

A few good heartnut trees have been located at various points. One of these trees is about thirty-five feet tall, with a spread of nearly sixty feet from tip to tip of branches. The present owner harvested several bushels of good nuts in one season from this tree.

I bought with my own funds a bushel of nuts from this tree and sent them in lots ranging from six to thirty to interested parties in various parts of Ontario. Of course I know that this is not in accordance with the best nut cultural principals, but I thought it was one way of getting nut trees started. If these nuts do not reproduce true to type, they will serve as a good stock for budding or grafting with the best introduced heartnuts later on. Another good heartnut was located almost on the outskirts of Toronto. At five years from planting this tree bore one-half bushel of fine, thin-shelled nuts.

In my last paper I stated that filberts had not done well in Ontario. I am glad to state that I will now have to retract that statement and inform you that good filbert trees have been found near Ancaster, which is close to Hamilton. These trees were about fifty years old, the largest specimen being nearly a foot in diameter at the base and about 25 feet tall. The trees bore well, but on account of the hordes of black and grey squirrels very few nuts were harvested. A fine lot of filberts was also found at Tyroconnell, a small hamlet on the north shore of Lake Erie, in Elgin County. These trees are nearly fifty years old and bear excellent nuts. Much to my surprise I found a fine clump of filberts growing quite near the campus of the O. A. C. at Guelph. These trees were introduced from England about sixteen years ago and at first they did not appear to be hardy, but eventually they established themselves and are now doing well in growth and fruitfulness. I was somewhat amused to think that I was searching so diligently for valuable nut trees all over the Province and did not even know of the existence of these trees, until a year and a half after I made my initial attempt to discover valuable nut trees.

I will have to correct another statement made at the last meeting, to the effect that almonds do not grow well in Canada except on Vancouver Island. Since then I have found a few, good, hard-shelled almond trees growing and yielding well in the Lake Erie country. This leads me to believe that almonds can be grown, with reasonable success, anywhere in the peach belt, particularly in the lake district.

In addition to my efforts to locate good trees I persuaded the authorities at the O. A. C. to establish small plantings of some of the best black walnuts, hickories, Japanese walnuts, and Chinese chestnuts. I also obtained about five bushels of Chinese walnuts and one bushel of Chinese chestnuts from northwest China for testing at the experiment stations, and by other interested individuals. Owing to the length of time the nuts were in transit the majority of them were unfit for germination. A few have grown, however, and we hope to get good results from these.

A collection of nuts containing 60 plates and 21 different species was prepared and exhibited at the Royal Winter Fair at Toronto and also at the Livestock Show at Guelph. I was in attendance almost constantly at Toronto, and endeavored to give all the information possible on nut culture. Both exhibits attracted a great deal of attention and called forth favorable comments from visitors and the press.

Experimental plantings of English, Japanese, Chinese, and American walnuts, filberts and hickories, have been established at the Horticultural Experiment Station. Mr. W. J. Strong pollenated about 200 black walnut blossoms with pollen of the English walnut. Apparently a good number (approximately 75%) have set fruit.

A graduate of the Ontario Agricultural College, who has become interested in nut culture, procured 2,000 black walnut seedlings from the Forestry Station at St. Williams. These trees were budded, in August last, with local grown English walnuts, but unfortunately only a few buds took. An attempt will be made next spring to whip graft the trees that did not set buds this summer.

There is a marked increase in the interest in nut culture shown by the public during the past year. This is shown by numerous requests for information and addresses on nut growing and by the public endorsement of nut culture by three important horticultural organizations. The Ontario Horticultural Council, the Federal Horticultural Council and the Ontario Horticultural Societies Convention each passed a resolution asking the Dominion Department of Agriculture to appoint a man to investigate the possibilities of nut culture in Canada. No definite action has been taken as yet, but it is expected that an appointment will be made in the near future.

We are giving the boys and girls of Ontario an opportunity to assist us in our work by hunting for good nut trees, and as an incentive we have offered prizes of $5.00 each for the best specimens of our various native and introduced nut trees. This should bring results, because if there is anyone in this wide world who knows where good nuts are, it is the small boy.

The work during the past year has generally been encouraging, but like every other line of human endeavor there have been disappointments. For example, one bushel of Chinese walnuts was stolen, and a number of good specimens of other kinds mysteriously disappeared from my exhibition collection.

Another disappointing feature has been the apathy, and even hostility, shown by some officials. I do not intend, however, to let these difficulties discourage me in the least, but plan to carry on and preach the gospel of beauty and utility as exemplified in our best nut trees.

ADDRESS BY DR. L. C. CORBETT

U. S. Department of Agriculture

The work in nut culture by the Department of Agriculture antedates the present Bureau of Plant Industry, and to confine the history of the work to the present Bureau of Plant Industry would not quite do the subject justice.

From the time of the beginning of fruit work in the Department of Agriculture, in 1885, nuts have received more or less attention. After the formation of the Bureau of Plant Industry, in 1901, special appropriations were received from Congress for the support of nut investigations, and individuals were appointed to that service in the department. Mr. C. A. Reed, whom you all know very well, was the first appointee of this service, devoting his whole time and attention to the work. He has been with the department for several years, and has given his time exclusively to the nut problems of the country. Naturally, the nut problems are not confined to any geographic area, but are nation-wide; but certain of the plants which have entered into the problems of nut culture have demanded more attention than others, for reasons that are the same as in fruit culture. The older fruits, those better known and longer in cultivation, whose problems are better understood, require less attention from the grower and from the experimenter than do the newer ones in the field.

Nut culture in America, as I understand it, not being a nut culturist myself, consists of two types of projects. We have one type that has long been practiced by man, that we imported from European countries and established on this continent. People have cultivated these nuts more or less intensively for generations, and many of the problems have been worked out, so far as Europe is concerned. Of course, when introduced in America, new problems confronted the growers here. The other type of nut industry is based upon indigenous nuts of which we know little, either from the orchard standpoint or as to the varieties concerned. Our native nuts, particularly the pecan, have forced themselves upon the attention of investigators of the department to much greater extent, perhaps, than any other nut with which we have to deal. Being a native, indigenous plant, not yet under cultivation, there is immediately presented the problem of the choice of varieties, adaption to changed conditions, and all of the problems arising in connection with a rapidly developing commercial industry; certain enthusiasts soon become enamored with the possibilities in the southern parts of the United States for pecan culture, and they immediately transplant it into new and untried regions, and as a result their problems have become legion.

The work of the Department of Agriculture in nut culture has developed really around the growing industries of the country; primarily, around the pecan, and secondly, around the almond and the walnut, for these are the more important, commercially. Naturally, the most pressing problems arise in connection with growing industries; they have growing pains which have to be eased the same as with small boys.

The Department of Agriculture has therefore found itself in the position of seeking answers to numerous questions which have been made in connection with these developing industries. I believe that we have contributed very materially to the knowledge of varieties, particularly as regards their adaptation to different geographic locations. We have also assisted the industries to solve some of their problems of cultivation, particularly of propagation, and also the problems growing out of the maintenance of soil fertility. With a new crop, in a new environment, it is always a problem to know how to manage the soil, and this is one of the leading lines of activity in the field, at the present time. In the Bureau of Plant Industry, two offices, that of Horticulture and Pomology and that of Soil Fertility, are co-operating in the solution of the soil fertility problems in the pecan regions.

Of course, as the industry developed and became established, the natural enemies of the pecan and of the other nut trees asserted themselves, as a result of which there have been set up investigations in the Bureau of Plant Industry to study the life histories of the various fungi that attack pecans; and outside of the Bureau of Plant Industry, the Bureau of Entomology has been devoting time to the study of the control of insect enemies. So that, at the present, the department is so organized that three or four important lines of attack are being made upon problems of these industries. Thus, while at the beginning of the Bureau of Plant Industry, in 1901, there was no single, individual person devoting his time and attention to the problems of nut culture, at present there are quite a group of individuals giving their whole time. I feel we are making progress in the work, and while we may be lagging very much behind what we should like to do, we are assisting as best we can, and are at least keeping in sight of the industry, as it goes forward.

I will not try to go into details about the work we are carrying on, because it is better to tell of what we have accomplished than to tell what we hope to do. We have a man on the Pacific Coast giving his whole time and attention to the study of breeding and of the cultural problems of almonds. Besides this, we have two men giving all of their time to pecans;

and during the last year, there has been established near Albany, Georgia, a station devoted to the cultural problems of pecans. One gentleman is continuously on the ground with the work, and two others devote more or less of their time to it.

Now, while these problems connected with the industries are the ones occupying most attention, the workers in the Department of Agriculture have not been unmindful of other native nut-bearing plants, such as the native black walnuts, the hickories and the chestnut up to the time of the very destructive attack of blight. The chestnut, however, has not passed out of our sphere of activity, because at the present time, (and I think you will see tomorrow at the Bell Station, some interesting possibilities in the future of chestnut culture in this country), the Chinese forms, which are much more resistant to blight, bid fair to give us a progeny to make it possible for us also to have a chestnut industry from the horticultural standpoint.

Probably the day of timber supply from our native chestnut is at an end. We hope not, but it looks that way at the present time. The possibilities of growing trees from China, the mollissima, or hybrids of them, bids fair to place the chestnut industry so that we can contend with the blight. We probably will not have immune varieties, but those which are able to live with the blight. That, it seems to me, is a very important consideration, because chestnuts have always been an important nut in our eastern markets, and are important in the European markets as well. While the larger forms of southern Europe will probably not be of value to us here, if we can establish a nut industry with nuts of fair quality, as large as our native sweet chestnuts, based on the Chinese species, the mollissima, then we will be making progress. You may see some of these trees at Bell Station which are eight or ten years old; they are bearing quite abundantly, and some of the chestnuts are really very palatable and of satisfactory size.

In addition to this breeding work with chestnuts, there is under way intensive breeding work with almonds which has for its object the development of those more hardy than those now in cultivation in California. This almond industry, though large, is handicapped because of the late frost injury, and it is desirable to get those which will bloom later and withstand lower temperatures.

The varietal problem with pecans will be ever with us, as long as varieties are found in the wilds and as long as people continue to plant seedlings in different localities. That is one of the subjects that is being given considerable attention.

In addition, the relative productivity of the plants to use as mother plants is an important one. In the work of the Department of Agriculture in connection with citrus fruits, it has been found that the individual bud carries over into its progeny the ability to produce fruit not only of a given type, but also the productivity of the parent to the progeny. A long series of records of the behavior of individual trees have been secured; we are building up a mass of information on which to base selections for better parent trees than any available at the present time. If the pecan behaves like the citrus fruits of California, we will be able in the future to have strains and varieties which will be very much less variable than those at the present time.

The propagation, selection, disease and cultural work covers the field that is handled by the Bureau of Plant Industry. We always like to dream of the future, and we are pleased to have the dreams come true. We must have in mind the possibility of better black walnuts than we have at present; and after the great inroad into the industry made at the time of the War, when the trees were used for timber purposes, there should be a greater effort on the part of the people in the northern districts to propagate black walnuts, not only for nuts but also for timber. The black walnut is a very great asset not only for timber and for ammunition purposes, but for food as well.

The hickory tree is in the same class as the black walnut—it is a valuable timber tree as well as nut tree. No other timber is as valuable for the construction of wheels as hickory, and while the "disc wheel" has served a useful purpose in railroad car construction, it is not likely that it will replace hickory altogether in the construction of wheels of motor vehicles. We are veritably a nation on wheels and we will always be looking for material with which to carry us through the country. As I have said, we are a nation of people on wheels, and if your propaganda did nothing more than to stimulate an increased interest in the production of hickory for timber purposes, it would be accomplishing a great result. But I believe that there are varieties among the hickories which should be to the North what the pecan is to the South. There are those which are very large and those which are thin-shelled, and those of fine flavor; as a food product I think the shellbark is second only to the pecan. And I should hail the day with great interest when there are good, recognized varieties of hickories corresponding with the best varieties of pecans. I believe they will be found and developed.

I have told you something of what we are doing and of what we hope may result. I hope that you will all visit the offices of the Department carrying on this work, and that you will get acquainted with the men handling the various projects, and tell them what your troubles are, that they may know how to proceed, and that they may discuss with you the best ways of attacking and handling the problems with which you are confronted.

Prof. Lumsden of the Federal Horticultural Board spoke of the chestnut bark disease and the fact that our experts advise us that within the period of twenty-five years the destruction of the native American chestnut will have been accomplished. The tanners and related interests of the country are now scouting around to find some species of tree to use as a substitute for tanning operations. Castanea mollissima is capable of developing into a good sized tree. From an economic standpoint the texture of its lumber is good, while the quality of its fruit is fair, and as an ornamental tree it has a future. It has resistance to the chestnut bark disease. It may become a substitute for C. dentata. Several crosses have been made between C. dentata and C. mollissima and some of them show considerable merit. Selection of these hybrids will have to be made for two purposes, namely wood production and fruit production.

Corylus colurna, the Constantinople filbert, is destined to become popular as an ornamental. On the Pacific Coast a bacterial blight occurs in some sections on corylus. A great work can be done in this country by the Northern Nut Growers Association by publishing bulletins advocating plantings of nut bearing trees for a three-fold purpose, timber, food, and beauty.

Communications were read from Miss Frances L. Stearns, Instructor in Botany of the Grand Rapids (Mich.) Junior Colony, asking information about planting nut trees, and from Mr. J. A. Young, Secretary of the Tree Lovers Association of America, asking the association to adopt their slogan and to co-operate with it in urging the more intelligent planting of trees, shrubs and flowers.

The evening session on Sept. 26th was called to order at 8:10 and a moving picture reel, "The Almond Industry in California," loaned by the Dept. of the Interior, was shown. Following that an address with lantern slides was given by Mr. C. A. Reed of the Dept. of Agriculture, on his recent trip to China.

MR. REED: In 1910 certain Americans in China conceived the idea of exporting the walnuts produced in that country to America. The experiment proved so successful that they continued to do so, and shipped their walnuts to this country year after year. The business built up very rapidly, until the war broke out when, for the time being, the industry was forced to a standstill. But as soon as the war was over the business picked up again, and had assumed such proportions, about two years ago, that American growers wanted to know how much longer the Chinese would be able to send walnuts over here. Most of the nuts from China were of inferior quality to those produced in this country. Records of the exports showed that there had been an increase from China each year; but as to the methods used, the extent of orcharding, or the growth in planting, etc., the matter had not been written up, and the consuls had not the remotest idea. It was finally decided by Congress, therefore, that a special appropriation for an investigation should be made. So a special trip was made to China to ascertain, first of all, the probable trade from there for the next ten or twenty years. Our people felt that more walnuts would be coming here, and they wanted to know about this before they planted any more here. It fell to my lot to make the trip, a year ago this summer.

We went first to Honolulu; then to Manila and Japan, and finally to China. We went into the section just to the right of Tientsin. By superimposing a map of China over that of the United States you may see that China more than covers this country; China is considerably larger than the United States.

Our basic point was Peking, which is in about the same latitude as Philadelphia. We found that walnuts were grown all through this section of China, not very much farther north than Peking, but not much farther south than Shanghai. There are walnuts cultivated here, in the Chinese way, over a great area; but we were convinced that the exportation of walnuts to this country was not likely to increase, for the business has apparently reached its height. American trade takes the best nuts; the second best go to Canada, the third to Europe and the fourth and fifth to Australia.

Our first expedition into the country was almost directly north of Peking. We went down the railroad about 15 miles, to Shaho, where we employed donkeys and a ricksha, and rode across country some 12 or 15 miles. Here we found a very excellent Chinese hotel, and surrounding orchards of perhaps 300 trees. Some of the consular reports in China stated that this place was one of the three sections in which the finest shipments of nuts were produced.

We next went to the east of Tientsin where we found quite a number of orchards and trees claimed to be from 150 to 200 years of age, although we found, after travelling a short time and inquiring from the Chinese farmers, that the figures they gave to us were probably inaccurate. We finally ceased to ask the Chinese farmers for figures of that sort. It was very interesting to note the difference in Chinese and American methods. For instance, in China, the land may be owned by one or by several people, who will lease the land or the trees, or perhaps even an individual tree, for a period of years. White marks placed on the trees indicate their ownership.

Young walnut trees were very scarce. We were told in one province that Chinese merchants, who had been forced out of Russia because of economic conditions there, and had lost everything, had come home and were seeking something with which to make money. They were already planting a considerable number of walnut trees, and were growing crops under the trees, planting crops of millet first, and then of soy beans later in the season. Another crop they use is called kaolin (pronounced "gollin" in this country).

Very few of the trees are ever pruned systematically, or taken care of; the Chinese seem to have no idea of this. Of course, the rainfall there is at a different time of the year than ours. Fall, winter and spring, in North China, are practically without rain. Consequently, the atmosphere is very dry.

Here and there we found trees that struck us so favorably that we made notes with the intention of going back to the trees to get scions for propagating purposes for this country. We were told that one of these trees had borne 800 pounds of nuts. I suppose, however, if that was so, it was green weight, and included the hulls. This tree was on the grounds of the Y. M. C. A., about 80 miles below Shanghai, the farthest south we went. The tree had been planted by missionaries, and had made splendid growth. There were not many walnuts south of that point, however. In the province of Shanshi the soil is of a washed nature, subjected to rains, and we found there huge gorges that had evidently been forming for centuries. All of the soil there, that is not too uneven to be cultivated, is terraced; and along the sides of the terraces walnut trees are planted. We usually found tunnels along the sides of the terraces. These were dug around the bank so that the water would run through the tunnels instead of over the terrace.

We saw no indications of blight. We thought we saw it in one case, but when we examined the nuts, it proved to be nothing but insects working on the hulls.

Wherever we went, we were told by the Chinese that they harvest their walnuts at about the time of the year which in America would be about the first week in September. We found, however, that the nuts were off of the trees and assembled on the ground for sorting and drying, long before that. They were put in windrows covered with millet straw and left for ten days, after which time the hulls were chipped off with knives and the nuts immediately washed and put on the market. I was particularly struck with the mechanical motion with which the Chinese men worked; it was just as regular as a machine. This was the first time that characteristic came to my attention, and afterwards I was struck with the same thing everywhere.

Each farmer takes his products, whatever they may be, to a common town called "market town," and there they are bought by the local merchants, or the "compradors." The exporters are missionaries and foreigners who make no effort to buy from the farmers, for the tradesman, or comprador, can get the nuts at a better figure than can the foreigners. The tradesman gets his commission in addition. The baskets of nuts are carried on poles placed over the shoulders of the Chinese.

One of the principal walnut centers of Chantung Province is 25 miles from the railroad, and we made quite an effort to reach it. An agricultural missionary, a Mr. Gordan, made the trip there with me, and we found it a badly infested section. We arrived about three o'clock in the afternoon and took about one hour going around to see the nuts. There were places within the wall where nuts had been assembled, and we made estimates as to the number of pounds. I think there were from 100 to 150 sacks of nuts in a pile.

Many of the women and children grow walnuts and these crops are inspected and sorted before being shipped to Peking. In the early summer, we saw quantities of apricot kernels being transported to the market and sold as almonds. We had understood that China was quite an important almond-producing country, but I doubt if there are any almonds in China. I did not see a tree, nor did I get an indication that there were any there.

One of the largest chestnut trees that I saw measured eight feet and would have been valuable for timber purposes. It was in one of the very attractive little orchards of chestnut trees in the north of Shansi and northeast of Tientsin. We understood that there were very large orchards to the north, but you might say that there is no such thing as a large orchard in China. We counted about 100 trees in such orchards, and we made notes as to their bearing habits. We found the chestnuts of pleasing quality, of a fair size, and not quite as large as European nuts but larger than the American. We did not see many of the trees which had been allowed to develop normally. They are not of special value in China, and consequently, the branches are removed as high as possible, and often the tops are cut out.

The Chinese have a species of native peanut which is very shrivelled and hard; but missionaries from this country have introduced there the American peanut, which is now grown so extensively that Chinese exports have disturbed our market conditions considerably.

The Chinese allow nothing to go to waste. When the peanuts are removed from the ground and cared for, the soil is sifted so that no peanuts will be lost. The American peanut grown there is served in little butterdishes on the hotel tables, as a delicacy.

THURSDAY MORNING SESSION, SEPTEMBER 27

Meeting called to order by President McGlennon, 10:15 a. m.

The president appointed as Nominating Committee to nominate officers for the ensuing year, Dr. Robert T. Morris, Prof. C. P. Close, J. S. McGlennon.

Mr. T. P. Littlepage, of Washington, D. C., then spoke on the subject of Commercial Nut Culture.

This is a very difficult subject to discuss, for the reason that, as yet, there are very few facts upon which to base any conclusions about commercial nut culture in the North.

First, let me say that the principal point upon which we base our opinion that nut culture in the North has commercial possibilities, is the fact that growing throughout many sections of the North are thousands of nut trees, pecans, walnuts, hickories and butternuts, many of which grow very fine nuts. It would be a repudiation of all known laws of natural science to conclude that trees budded and grafted from these desirable parents would not grow and bear the same as they do. Therefore, we are perfectly safe in concluding that if there are successful nut trees growing, others also will grow. Let us proceed to consider some of the requirements.

First, there is the soil requirement. But before considering the soil requirement, I might add that we must keep within reasonable latitude of the homes of the native trees. This subject has been fully covered in previous reports of our association, and I do not care to go into a detailed discussion of it, except to say that prospective planters of commercial orchards should read the previous reports of the association on this subject, and keep in mind that somewhere north of the home of the parent trees, is a line north of which these trees will not bear. This line is dependent upon several things, altitude, topography and other elements. As an example, I merely mention that orange orchards flourish in California at the Philadelphia latitude.

Going on with the question of soil, upon this subject alone might be written a whole volume. But a few points are essential. Most nut trees require a deep, well-drained soil that is not swampy or seepy, and over which there are no overflows during the summer season. Pecans grow along the river bottoms where there are heavy overflows in the winter, but such an overflow in the summer would probably kill the trees. Nut trees seem to flourish well on land that is underlaid with clay as a subsoil. In fact, almost any kind of good farm land is suitable for some of the different kinds of nut trees, provided it does not come within the restrictions above mentioned. The better the land, however, the more successful will be the growth of the trees, and I very much doubt whether it pays to put any kind of desirable tree on undesirable land. I have heard it said of pedigreed stock that about ninety percent of the pedigree is in the corn crib, five percent in the man that does the feeding, and five percent in the blood. Perhaps these percentages might be subject to some variations. I shouldn't reduce the corn crib requirement, and I think about ninety percent of the success of our nut trees will depend upon the land.

The next point to be considered is the question of varieties and, in this connection, it is essential to remember that nuts are produced to be sold and eaten; therefore, it is important to keep in mind the requirements of the consuming public. Upon this question also have been written many thousands of pages which, when all summed up, simply amounts to this: get the best varieties that will bear in your particular locality. This can be determined to some extent by what native trees are growing in your particular locality, although not entirely so. In many sections of the country, there are no native pecan trees, and yet these trees flourish very successfully when brought from some other section. On this point the prospective planter of commercial orchards should seek the best advice obtainable.

The third requirement for a commercial nut orchard is cultivation and attention. Many of the nut trees will grow and bear without any attention whatsoever, but they will take your time for it. I have seen wild pecan trees that were not over twelve or fifteen feet high at twenty-five years of age. I have seen cultivated trees larger than that at eight years of age. A tree responds to care and cultivation the same as corn or potatoes or any other of the cultivated crops. The lack of cultivation is just as detrimental to them as to these crops. Young pecan trees should be hoed five or six times each summer, and when they get to be four to seven years of age, there ought to be a constant, clean cultivation, from early spring until late in the summer, followed by a good cover crop to be turned under the following spring at the beginning of the cultivating period. They should also be given plenty of good, commercial fertilizer.

If the prospective planter of commercial nut orchard has enough faith and hope and follows the suggestions given above, he will not be dependent upon charity in his old age.

DR. JORDAN: I am interested as an amateur pecan grower, and I would like to ask what varieties will be of most profit, commercially, that can be grown with a reasonable hope of success in the northern latitude.

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MR. LITTLEPAGE: The question is a very difficult one to answer, but the important thing is to stick to the kind that grows the best in your locality. The Posey is grown in Lancaster County, Pa. The parent Posey tree grows in Indiana, and I had the pleasure of naming it. That tree is a good bearer, and it is the thinnest-shelled northern-grown pecan with which I am familiar. It is a very beautiful nut, with the exception that frequently one side of the kernel will not fill out as it does on the other sides. It is not defective, but simply deficient. It will have one full sized kernel but it is not perfect in shape. I myself do not think this a very serious objection.

The Major is a fine bearing pecan, but the question is whether it is large enough to be good commercially. The Niblack is the highest flavored pecan.

The following letter from Mr. J. F. Jones, vice-president of the association, was then read:

I am very sorry not to be able to attend the meeting this year. My son, who has the overseeing of the outside work and, in my absence, the general work, is incapacitated, due to an operation for appendicitis last week and, with a number of men at work on particular jobs, I cannot get away.

I am sending a few nuts which may be of interest to visitors. About half of my young pecan trees are bearing this year and a few trees are quite full. So far, Busseron shows up the best in bearing, with Posey second, and Niblack third. The English walnuts are a good crop. Mr. Bush has a big crop of these, and older trees in general have a good crop. The Rush hazel is bearing a big crop as usual. So far this is the only variety in any species to bear heavy annual crops here. The weather, seemingly, has no effect on the setting of the nuts. Last spring we had it down to 10 above zero when this was in bloom, but it set a full crop from both hand and natural pollenization. Hybrids of this and the best large fruited Europeans which have come into bearing are very promising, but it is too early to judge as to their bearing.

Put me down for new memberships or cash as last year, or for my part in any arrangement that may be decided upon to take care of the indebtedness of the association, or to advance its usefulness. I shall also be glad to extend the offer of two nut trees as last year, to new members, if it is thought this will help in securing the new members. Offerings this year would be Stabler black walnut seedlings, Chinese, Mayette, Franquette, Eureka, etc., in the English or Persians. Also seedlings of the Rush hazel, if wanted.

Having been nominated vice-president of the association two years ago, it may be understood that I am in line for the presidency this year upon the retirement of our honorable president Mr. McGlennon. If so, I wish to ask the nominating committee not to consider my name as I cannot accept this responsibility. With the vast amount of correspondence incidental to supplying information to those wanting to engage in the growing of nuts or nut trees, and growing and selling nut trees, experimental work and breeding new types and varieties, I have my hands full and could not do this position justice. We also have members in the association better fitted for this position who can give it better thought and attention, and who can advance the association and the interests of nut growers more than I can, while I can be of more benefit to the association and the nut industry in general without taking on the duties imposed by any official position.

NOTES BY MR. BIXBY

Thursday, Sept. 27

Trip by automobiles to Mr. Littlepage's farm at Bowie, Md., and to the U. S. Experiment Station at Bell.

Mr. Littlepage has an orchard of 275 trees covering thirty acres of pecans and Stabler black walnuts, the first pecan trees being set in 1914, and the Stabler black walnuts some three years later. Now both are starting to bear, a few nuts having appeared last year, and a very few nuts the year before.

The trees are growing finely, the leaves have a fine dark green color, and nuts were noticed in clusters, the pecans being in clusters of 2, 3, 4 and 5; and the black walnuts in ones and twos.

That the orchard has been given good care is evident. Commercial fertilizers and green manures have been used. A winter cover crop of rye was grown last fall and plowed under this spring, and a summer cover crop of soy beans was grown this summer and will be plowed under this fall.

The varieties noticed in bearing were the Major, the Greenriver, Stuart, Busseron and the Indiana. Of the above, all are northern varieties, excepting the Stuart, which is a southern variety which has given evidence elsewhere of being able to grow and to bear further north than almost any other southern variety.

The pecans are set in blocks, the earlier ones being set 60' x 60'. Mr. Littlepage became convinced after his first plantings that this was too close, and the last planting of pecans was 100' x 120'.

The black walnuts are planted along two fence rows, the trees being fifty feet apart, the total length of the rows being about three-quarters of a mile. The peculiarity of the Stabler black walnut of bearing some nuts where the kernel is in one piece, that is where one lobe of the kernel has not developed, was noticed in some of Mr. Littlepage's trees. There is going to be, in future years at Mr. Littlepage's place, an opportunity to study this peculiar behavior of the Stabler black walnut, that could be carried on at the parent tree only with great difficulty, because of the inaccessibility of the tree, in the first place, and the inaccessibility of the flowers, owing to their great height above the ground, in the second.

At Bell Station was seen Dr. Van Fleet's work on chestnuts. Some ten years ago Dr. Van Fleet began this work for the purpose of getting something that should be blight proof, or at least strongly blight resisting and that would furnish the nuts which the chestnut blight is rapidly making impossible of production. With this end in view, some ten years ago Dr. Van Fleet planted nuts of the Chinese chestnut, Castanea mollissima, and planted out the seedlings. He also procured from the place of J. W. Killen, at Fenton, Md., nuts of Japan chestnuts that had withstood the blight up to the time the nuts were planted. The first thing to be found out was how well these would resist the blight. None were found to be immune, although the trees are still alive after ten years exposure. Dr. Van Fleet's ambition was to get a blight-resistant chestnut the size of the Japan chestnut with the delicious flavor of the chinkapin. This, as yet, has not been accomplished, although some very good nuts much larger than chinkapins were seen. One interesting fact noted as to resistance was that the Japan chestnut, which is not generally supposed to be as resistant as the Chinese chestnut, was at Bell Station apparently standing up just as well.

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At the evening session, Thursday, Sept. 27, a rising vote of thanks was given to Mr. and Mrs. Littlepage for their hospitality of the afternoon. The president then introduced Mrs. W. N. Hutt, editor of the Progressive Farm Woman, of North Carolina.

Mrs. Hutt quoted H. G. Wells as saying, "The primeval savage was both herbivorous and carnivorous. He had for food hazel nuts, beech nuts, sweet chestnuts, earth nuts and acorns." She went on to say:

In Spain and Southern France, the chestnut is now used much more than in the past. You should know in what appetizing forms they are cooked. It is a question how you should cook the chestnut if you do not want to spoil its flavor. Should you steam it, boil it, or what? When you want it in bread, or when you use the tasteless forms, it is first steamed or boiled, and later is mashed up and made into bread, or mixed with cheese or tomatoes. But if you want to develop the flavor, then roast it, pick it out from the shell and crush it, using almost no other flavor with it.

Have you ever realized how much we depend on the walnut in cooking? Take the pecan, or perhaps almost all of the nuts; the flavor is diminished by cooking. But the walnut is the one nut that gains in flavor by being cooked. This means a great deal for the popularity of the walnut.

A friend of mine was captured by the Germans, and was sent out each day into the forests to gather acorns to be used in the prisoners' food. The friend said that many a time he thought he would rather die than to have to eat or gather any more acorns.

Farmers' Bulletin No. 712, "The School Lunch," by Caroline Hunt, has been especially valuable in the preparation of the school lunch with nuts. There is a man who comes to North Carolina every winter, who will tell you that he lives on ten types of nut oils and nut butter.

The great mass of people out through the country are not yet ready to comprehend this; but once they are educated to the value of nuts, the demand for them will be unlimited.

As to the question of economy, the prices should not go up any farther; they will not be used enough until they become cheaper. With many boys and girls in a family, a dollar's worth of nuts, at $1 a pound, will not go far. If we could get nuts at more reasonable prices it seems to me that women would consider them more than they do for food. They want them not only for their parties, but in everyday life.

We should popularize nuts through newspapers. It pays to advertise, and little notices in the paper are much more far-reaching than any other way of telling the story of the nourishment to be found in nuts.

As to the value of nut trees in landscape work, a real estate man told me that when he wanted a good price for a house he planted fruit trees at the back of the house, and nut trees on the sides. He would talk about those trees to the people who came to buy, and has sold many houses in this way.

Then take Arbor Day, and we have one in nearly every state in the Union. If we could get the papers and the forest magazines to talk about Arbor Day, and urge everybody to plant something, and particularly to plant a nut tree, it would not be long before we got results. I could not think of anything much more patriotic than planting avenues of memorial nut trees. Nut trees are better to look at than are many of the monuments erected, and the patriotic societies do not realize the truth in this. There is a case where with a stroke of the pen, the nut trees could be increased all over the country.

Then consider the home demonstration agents in the country. They have the women organized and are in touch with the men of progressive thought and feeling everywhere; and it seems to me that we could make more use of them. It would seem that if this organization could in some way raise the money to have someone talk at these demonstration meetings, it would not be long before the value and the beauty of nut trees would show the use of doing this splendid work. What more effective methods could there be than to go to the state meetings held by home demonstration agents twice a year, and talk nuts to those people? They go home and talk these same things to all of the women in their little organizations and communities. There is no rapid transit method more effective than that. Then, when the women are taking up a subject like that, men are apt to read it also.

Another form of advertising that is equally important is in men's organizations. A number of years ago Mr. Hutt went down through the eastern part of the state on the old farmers' institute work. He took with him a case fixed up to display nuts. He talked about them, and especially about pecans. The people had never seen anything but the little, old, wild pecan, and they became enthusiastic. When you get a farmer enthusiastic you are doing something. The people became quite enthusiastic and planted quite a number of orchards. Mr. Hutt left the department and the new man who came in was not particularly enthusiastic about nuts. Then Mr. Curran came into the work and decided there was nothing he could do better than to urge them to plant nut trees. He is trying to get an unlimited quantity of pecans and walnut trees planted and he hopes to have a large number of trees put in within a few years.

To paraphrase what Mr. Littlepage said this morning, in connection with the raising of hogs, in getting the world to plant more trees, to use more nuts and to appreciate the value of nut trees for both beauty and use, you need 90 percent of advertising; and let the 8 percent be the man and 2 percent be the nut.

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DR. MORRIS: Last year, when my experiments with the use of paraffin grafting had apparently been completed, I included what I knew of this subject in a little book, and this brought out letters from all parts of the country, in fact from all parts of the world, reminding me that I had not completed the subject of the use of paraffin in grafting. From tropical countries men complained that my suggestions about the use of one particular kind of paraffin, "Parowax," were not applicable to their part of the country where the paraffin would melt in the summer sun. Then, from some of the regions where the nights were cold, they said the paraffin would crack and leave the stocks bare, owing to the change of temperature.

We are consequently faced with a necessity for extending our information on this subject. My reason for presenting it, before I have completed investigations, is to get suggestions from members of the audience here, and from practical nurserymen. I have written a number of books on various topics, and have never sent one out without feeling sorry that it was not time for the next edition.

The theory is that if we cover a graft completely with melted paraffin, including the entire scion, buds and all, we have accomplished several things. In the first place, the paraffin prevents the graft from drying out before new cells can make union with cells of the scion.

In the second place it fills all interstices where sap would collect.

In the third place it provides an airtight covering so that the free sap pressures, negative and positive, under different temperatures, will be analogous in stock and scion. When there is low sap pressure we assume that some of the sap may be drawn out of the scion. This airtight covering prevents that.

In the fourth place it provides a translucent covering, which allows action by the actinic rays of light, which brings the chlorophyll into activity. All plant growth is conducted under the influence of chlorophyll, and the actinic rays of light activate this. Consequently, I seemed to have a perfect grafting material in this Parowax, which we may find in any grocery store. In my locality this wax worked perfectly and, theoretically, nothing more was to be desired. It melts at 125 degrees farenheit.

I have brought with me a specimen of a pear tree that I grafted in this way in July of this year. You will see that the Parowax covering is still complete. The new shoots have grown about eight inches since July 1, and I do not see how you could imagine anything more perfect than this specimen, from which I wrote my description in the book. As a matter of fact it is by the use of the paraffin method that I seemed to have solved the very great problem of making it possible for anybody to graft anything, and at any time of the year. The most difficult thing to graft is the shagbark hickory, and we have even done that every month of the year, except December and January. This year we are going to try those months, for I believe that the hickory tree may be grafted any month of the year.

Now the point of my remarks will relate to different kinds of paraffin. This Parowax, which melts at 125 degrees farenheit, will be satisfactory in the north temperate regions. We may raise the melting point ten degrees, if we like, by the addition of the carnauba wax, which, however, is highly crystalline. A crystalline wax is not desirable because it cracks and permits the air to enter and we have a desiccation of the scion. The Standard Oil people will furnish paraffin with a melting point of 138 degrees, and that will cover all of our needs for hot countries. But in getting paraffins that melt at 136, 137 or 138 degrees we have a rather definite crystalline element. Mr. Bixby has suggested the use of the earth wax which is mined in Australia. It is really a fossil paraffin and is not so granular. I found that it is not to be had in this country at the present time, however, although various dealers told me that they had it, and I obtained from a firm in New York City a misbranded specimen called "Ozokerite," which they said is a technical term for this particular fossil paraffin. But it was nothing of the sort; it was something they had made up for themselves. Mr. Bixby kindly gave me a pound or so of the real "Ozokerite," so I had the genuine thing to experiment with. We may then settle the question of obtaining paraffines which have a high melting point, by knowing that they may be obtained from any of the Standard Oil people.

Knowing that we must have, in addition, the elastic feature, I found one man who had succeeded by adding something to a high melting-point paraffin. He said that it was a secret, but I soon found that it would be no secret to a bee. It would seem, then, that this quality in beeswax would be valuable, since the secret formula from this same dealer has little more than beeswax in it. Beeswax is a different kind of organic product from paraffin and I would not expect them to mingle naturally when in melted solution, but apparently they do. You will find that the specimens which contain this wax are very smooth to the touch, and apparently are more homogeneous than paraffin.

The subject for experiment then, for members of this audience, is that of finding some substance that may be added to give elasticity, but which will not change the melting point. In the South we may require in addition something to whiten our paraffin. Some men in Southern California wrote me that they had fastened white paper about each graft and put a rubber band over it. I suggested this plan to one or two men in Australia and in Ceylon, who had complained about the melting of the Parowax, and I have not yet received their replies. I have been trying, however, to simplify things in the way of grafting. In addition to the elasticity that we need, we must have whitening, and for this purpose we must add something that will not be poisonous to the tree but will mix with the paraffin readily and give a white paraffin, which will interfere somewhat with the actinic light. I have found that carbonate of lead will mix well with paraffin. Carbonate of zinc will also mix well. They are both heavy, so heavy that they need a certain amount of stirring. A lighter substance is citrate of zinc, which will give elasticity, and which will probably also give a white effect. It melts with the paraffin and, being neutral, it will do no harm to the tree.

I have given you an outline on which I wish discussion, for I hope to get from this audience the information and suggestions that will enable me to make my experiments in the right way so that by next spring we may have no further need for discussing the question as to the correct paraffin method in grafting.

MR. BIXBY: There is another wax that is not so crystalline as the Parowax, and that is Candelilla, which is produced in Texas and New Mexico. It may be obtained from the wax importers in New York City, not from the Standard Oil Co., but the importers. I will find out just where it is from. I can easily get samples. Its melting point is not so high as Parowax, but it is much higher than any of the other waxes.

DR. MORRIS: Then by mixing it with the high-melting point waxes, those of about 138 degrees, we might get good results.

MR. BIXBY: I think so, and without introducing the crystalline element.

Prof. H. H. Hume of Glen St. Mary, Florida was then asked to speak. He said that he uses fresh pine gum from the turpentine cups to make grafting wax stick. This will mix with beeswax and give the elasticity needed for winter work (in the South). Also it is unaffected by a temperature as high as 120 degrees. He uses a mixture of high grade rosin, beeswax and pine gum with which pieces of cloth are saturated. Gum should be obtained in the spring when it is purest. It is thin enough to pour out.

Dr. Zimmerman said that he had tried pine gum with paraffine and it would not mix.

Prof. Hume said that beeswax can be had in various shades up to pure white.

Dr. Morris said that black grafting wax attracts heat and excludes actinic rays. He prefers a translucent wax.

Prof. Hume stated that in the country where Jacksonville, Florida, is there are 100 miles of roadway under construction which will be planted with nut trees where possible. He added that once when he was ill for a long time the doctor finally ordered a glassful of milk and a handful of pecan kernels for his diet. He tried it and it worked.

Dr. Zimmerman said that for grafting wax he had used equal parts of paraffin, stearic acid and beeswax with good results.

Dr. Morris stated his belief that the simple splice graft is the strongest kind.

FRIDAY MORNING SESSION

Sept. 28th.

The chairman of the Committee on Incorporation was called upon for a report and spoke as follows:

MR. LITTLEPAGE: Under the Code of the District of Columbia there is a provision of law whereby any educational, scientific or charitable association can be incorporated and become a body corporate with all of the rights of any other corporation, so far as the corporate entity and liability is concerned. The provision of the District Code is a very liberal one and drafted to encourage such societies as this. The committee therefore thought it better to incorporate under this provision of the law than under that of some other state.

The advantages of incorporating a society of this kind are several. It makes the action of the organization that of a legalized corporation and takes away liability of individual members. If anyone should desire to donate money to the organization, we would have a corporate entity that would be responsible under the law for the safe handling of such funds. Under the law we can hold such funds up to the point where the income is not more than $25,000 a year. In the District of Columbia a corporation can take title to real estate, transfer property and do all necessary things in accordance with its by-laws. We therefore concluded that there could be no objection to incorporating under such laws. So with the consent of the other members of the committee, I prepared in my office the proper certificate of incorporation which, under the requirements of the Code of the District, are as follows:

> KNOW ALL MEN BY THESE PRESENTS, That we, the undersigned, all of whom are citizens of the United States and a majority of whom are residents of the District of Columbia, desiring to associate ourselves for scientific and educational purposes and for mutual improvement; and to organize a corporation under sub-chapter three (3) of the Incorporation Laws of the District of Columbia, as provided in the Code of Law of the District of Columbia, enacted by Congress and approved by the President of the United States, do hereby certify:

FIRST: That the corporate name of this company shall be The Northern Nut Growers Association, Incorporated.

SECOND: The term for which is it organized is perpetual.

> THIRD: The particular business and objects of the society are the promotion of interest in nut-bearing plants, their products and their culture, and, in general, to do and to perform every lawful act and thing necessary or expedient to be done or performed for the efficient conduct of said business as authorized by the laws of Congress, and to have and to exercise all the powers conferred by the laws of the District of Columbia upon corporations under said sub-chapter three (3) of the Incorporation Laws of the District of Columbia.

FOURTH: The number of directors of the said corporation for the first year of its existence shall be five.

IN WITNESS WHEREOF we have hereunto affixed our hands and seals
this 27th day of September A. D. 1923.

Karl W. Greene (Seal)
Albert R. Williams (Seal).
Thomas P. Littlepage (Seal).

DISTRICT OF COLUMBIA, TO WIT:

I, Alice B. Watt, a Notary Public in and for the District aforesaid, do hereby certify that Karl W. Greene (of the District of Columbia), Albert R. Williams (of the District of Columbia) and Thomas P. Littlepage (of the State of Maryland), parties to the foregoing and annexed certificate of Incorporation of *THE NORTHERN NUT GROWERS ASSOCIATION*, *INCORPORATED*, bearing date on the 27th day of September, 1923, personally appeared before me in the District aforesaid the said Karl W. Greene, Albert R. Williams and Thomas P. Littlepage, being personally known to me to be the persons who made and signed the said certificate and severally acknowledged the same to be their act and deed for the purposes therein set forth.

WITNESS my hand and seal this 27th day of September, 1923.

ALICE R. WATT,
Notary Public.

My commission expires December 17, 1923.

The smallest number of members with which corporation is possible, is three; so I secured two members, Mr. Greene and Mr. Williams, who, together with myself, prepared this, and put it in proper form. We then filed it with the Recorder of Deeds, keeping a copy for the files of the incorporation. The Recorder received it, and the fact that he received it was proof that it was satisfactory. We are now, therefore, a corporation.

Of course, we want to put that machinery into action, but in order to do so a board of directors has to be selected. Then will follow the election of officers of the Association. Therefore, I have prepared a report of the meeting of the incorporators, which I will read. As I said, however, we did this to get the machinery into operation. Next year the directors will be elected by the members.

MEETING OF THE INCORPORATORS OF THE NORTHERN NUT GROWERS ASSOCIATION, INCORPORATED.

The organization meeting of the Incorporators of the Northern Nut Growers Association, Incorporated, was held at Washington, D. C., September 28th, 1923, at 10:00 o'clock a. m.

Present: Karl W. Greene, Albert R. Williams, and Thomas P. Littlepage.

Upon motion, Thomas P. Littlepage became Chairman of the meeting.

Upon motion of Mr. Greene, seconded by Mr. Williams and unanimously passed, the following were elected Directors of the Northern Nut Growers Association, Incorporated, for the first year of its existence or thereafter until the annual meeting of the company in 1924.

James S. McGlennon, of Rochester, New York.
W. C. Deming, of Hartford, Connecticut.
Willard G. Bixby, of Baldwin, Nassau Co., N. Y.
Harry R. Weber, of Cincinnati, Ohio.
Robert T. Morris, of New York, N. Y.

Upon motion of Mr. Greene seconded by Mr. Williams and unanimously passed, by-laws of the corporation were adopted.

There being no further business, the meeting of the Incorporators adjourned.

KARL W. GREENE,
ALBERT R. WILLIAMS,
THOMAS P. LITTLEPAGE,
Incorporators.

THE PRESIDENT: The next action, then, Mr. Littlepage, would be to get the report of the nominating committee. I call for that now.

Mr. Littlepage: (Reads as follows):

MINUTES OF FIRST MEETING OF DIRECTORS OF THE NORTHERN NUT GROWERS ASSOCIATION, INC.

The first meeting of the Directors of the Northern Nut Growers Association, Incorporated, was held at Washington, D. C., September 28th, 1923.

Present: James S. McGlennon, Willard G. Bixby, Robert T. Morris.

Upon motion of Mr. Bixby seconded and unanimously passed, the following officers were elected for the ensuing year, or thereafter until the annual meeting of the Incorporation to be held in 1924:

President, Harry R. Weber; Vice-President, J. F. Jones; Treasurer, H. J. Hilliard; Secretary, W. C. Deming.

There being no further business, the meeting adjourned.

WILLARD G. BIXBY,

Secretary of Directors' Meeting.

(The report was adopted by the convention).

REPORT OF THE FINANCE COMMITTEE

By Willard G. Bixby

MR. BIXBY: The finance committee asks the association to instruct the secretary in the printing of the next report to endeavor to reduce the size to one-half of the present report.

(Adopted by the convention).

MR. BIXBY: I move as an amendment to Article Two of the By-Laws, that annual membership be $3, or $5 including a year's subscription to the Journal. Contributing members to pay $10, this including a year's subscription to the Journal.

(Motion seconded and adopted by the convention, and the committee on Incorporation discharged with the thanks of the association).

MR. LITTLEPAGE: I have nearly overlooked the fact that the organization must now have a corporate seal, with an appropriate inscription. An appropriate inscription would be "The Northern Nut Growers' Association, Incorporated." All such seals generally carry some appropriate design, and there are various ones to be had. I move that a committee of three be appointed to determine upon the design of this seal, and then later, if the chairman of the committee will send the design to me, I will have the seal made and send it to the association.

(Motion seconded and adopted, and Dr. Deming, Mr. Bixby, and Dr. Morris appointed as committee by the president).

After considerable discussion New York City was selected as the place for the next convention and the dates Wednesday, Thursday and Friday, September 3rd, 4th and 5th, 1924.

A vote of thanks to the president, Mr. James S. McGlennon, was adopted. The secretary was also instructed to write to Mrs. Hutt expressing the thanks of the convention for her address.

Dr. Oswald Schreiner of the Bureau of Plant Industry, U. S. Dept. of Agriculture was then introduced and spoke as follows:

In the successful growing of pecan trees, the proper care of the orchard is of enormous importance. (To illustrate this point, slides were shown of a good orchard and a poor orchard on a rather thin soil in the Coastal Plain Region. In the good orchard, the trees had been well cared for, the soil fertilized by the growing of legumes and cover crops plowed under; in the poor orchard, the trees had been neglected and the soil impoverished by the continuous growing of cultivated crops, such as cotton and corn. The two views very clearly showed which orchard was on a paying basis and likely to prove a profitable investment). It is needless to say that the crop from such a poor, intercropped orchard would be meagre and unprofitable until the methods were changed. The growing of legumes to furnish humus, and even the growing of winter cover crops, such as rye, to be plowed under in the spring, cannot be too strongly recommended as soil improvers.

When nut trees are grown in orchards, they can no longer be considered as forest trees to be left to take care of themselves until a rich harvest of nuts is produced, but must be cared for just as much as any other fruit tree or cultivated crop or the harvest of nuts will never be forthcoming.

The fertilizing of nut trees, however, offers more difficulties than do the annual crops. Experiments on this subject have been few and the information obtainable is rather meagre. Consequently, a few years ago, the Office of Soil Fertility Investigation, which is conducting fertilizer investigations on a large number of the annual crops grown on the prominent soil types or soil regions of the United States, started, in co-operation with the Office of Horticultural Investigations of the Bureau of Plant Industry, a number of fertilizer experiments on pecan orchards, involving a study of several soil types suitable for nut production and attempting to ascertain the proper fertilizer requirements for the pecan on these soils. While these experiments have been running only five years, which in point of time is very small in the life of a pecan tree, yet the different fertilizers employed already show some highly interesting results, sufficient to indicate that certain fertilizer applications undoubtedly influence the growth of the tree, its productiveness, and quality of the nut produced.

The experimental fertilizer mixtures are all prepared here in Washington in a fertilizer-mixing plant on the department's Arlington Farm, on the Virginia side of the river. The fertilizer house is well stocked with all of the various fertilizer substances used in agriculture, ready for mixing; nitrate of soda from Chili, potash from France and Germany, and our own far western states; cottonseed meal from the South, tankage and dried blood from the slaughter houses of Chicago and Omaha, Tennessee or Florida phosphates, and acid phosphate, ammonium sulfate from the coke ovens of

Pennsylvania, Thomas slag from England, in short, all sorts of commercial materials from near and remote sources, for study and use in fertilizers.

(Slides were then shown of the exterior and interior of the plant where literally thousands of experimental fertilizer mixtures are prepared to study the requirements of the various soils and crops, and are then shipped in freight cars to the various experiment places. Two slides showing the application of fertilizer in a large orchard where tractors are employed in carrying on the various cultural operations and also in a small orchard where hand labor is employed, were also shown).

The scheme of fertilizer experimentation adopted in this work is rather complete and so planned as to include fertilizers carrying the principal fertilizer constituents, phosphate, ammonia and potash, singly, in combinations of two elements, and in combinations of three elements, in various proportions in a regularly graded manner. The following scheme illustrates these mixtures of different analyses, the first figure denoting the percentage of phosphate, the second the percentage of ammonia, and the third the percentage of potash in the fertilizer. The various mixtures are numbered consecutively.

1

—-

20-0-0

2 3

—- —-

16-0-4 16-4-0

4 5 6

—- —- —-

12-0-8 12-4-4 12-8-0

7 8 9 10

—- —- —- —-

8-0-12 8-4-8 8-8-4 8-12-0

11 12 13 14 15

—- —- —- —- —-

4-0-16 4-4-12 4-8-8 4-12-4 4-16-0

16 17 18 19 20 21

—- —- —- —- —- —-

0-0-20 0-4-16 0-8-12 0-12-8 0-16-4 0-20-0

It is quite apparent that in this scheme the entire field of fertilizer formulas is covered in a regular way. In addition to this formula plan other experiments are also under way to determine the influence of the different fertilizing materials, carrying the phosphate, ammonia and potash, and the influence of lime, rock phosphate, various green manuring crops, etc. The experiments are carried out in commercial orchards on several soil types and in several localities.

While the years the experiments have been running are yet too few for any final conclusions, and the details too numerous to present in a brief sketch here, there have nevertheless been some very interesting results from the use of fertilizers which is readily shown by a few lantern slides. Here is, for instance, a view of a fertilized and an unfertilized section of one of our experiments in Georgia. The views were obtained in the fall, and one could tell at a glance, not only that the unfertilized trees were not as large, but also quite strikingly that they had nearly lost all of their foliage, whereas the trees on the fertilized section were still in full foliage, thus presenting a very strong contrast. The effect of fertilizers on the foliage is shown also in a series of slides of representative trees, from one of our experiments in Louisiana, likewise taken in the fall. The first tree had not been fertilized, the second had been fertilized with phosphate and the third with potash. The one fertilized with phosphate appeared slightly larger, but it can again be observed that all three trees were, at the time the picture was taken, nearly three-fourths defoliated. The next two trees from the same experiment, fertilized respectively with a nitrogenous fertilizer and with a complete fertilizer, and photographed at the same time, show the influence of these fertilizers strikingly in that they are still in complete foliage, as well as showing a more vigorous growth. Three slides of fertilized and unfertilized trees from still different experiments all show the fuller foliage and better branching of the fertilized trees, especially those fertilized with the nitrogenous fertilizers or the complete fertilizers.

The yields of these trees cannot here be taken up but, in general, these fertilized trees came into bearing earlier and have yielded double and treble the number of nuts produced by the unfertilized trees.

(In conclusion, there was shown a slide of the yield of nuts from an experimental tract of a commercial orchard of about 20 acres, in which the yield from a fertilized acre was compared with the yield from an unfertilized acre. It was noted that the unfertilized acre gave a yield of approximately two barrels, whereas the fertilized acre gave an increase of two bushel baskets more than the unfertilized.)

Dr. W. E. Safford, Botanist, Bureau of Plant Industry, then spoke on the Use of Nuts by the Aboriginal Americans.

DR. SAFFORD: My interest in nuts has been confined almost entirely to those of American origin. For a good many years, I have been studying the plants, and plant products, utilized for food, and for other purposes, by the aboriginal Americans, before the arrival in this hemisphere of Columbus and his companions.

In this connection, there is a striking contrast between the American Indians and the primitive Polynesians. The chief economic plants encountered by early explorers on the islands of the Pacific Ocean were identical with well known Asiatic species. Coconuts, breadfruit, taro, sugar cane, yams and bananas, the most important food staples of the Polynesians, had been known to the Old World for centuries before the Pacific Islands were visited by Europeans; the shrub, from the bark of which the Polynesians made their tapa cloth, was identical with the paper mulberry of China and Japan; and the principal screwpine, or Pandanus, from which the Polynesians made their mats, was a well-known species of southern Asia. A number of these plants had even carried their Asiatic names with them to Polynesia. The Polynesian language itself, with its varied dialects, spoken in Hawaii, Samoa, New Zealand, Easter Island and on other island groups, can be traced without difficulty to the Malay Archipelago, the cradle of the Polynesian race.

In America, on the other hand, every cultivated plant encountered by Columbus and his companions was new. Not a single Old World food crop had found its way to our hemisphere before the Discovery; not a grain of wheat, rye, oats, or barley; no peas, cabbage, beets, turnips, watermelon, musk-melon, egg-plant, or other Old World vegetable; no apple, quince, pear, peach, plum, orange, lemon, mango, or other Old World fruit, had reached America. Even the cotton which was encountered in the West Indies by Columbus the very morning after the Discovery, proved to be a distinct species and could not be made to hybridize with Old World cottons. Conversely, no American cultivated plants; no maize, no beans, tomatoes, potatoes, sweet potatoes; no cacao (from which chocolate is made); no pine-apples, avocadoes, custard apples nor guavas; no Brazil nuts, pecans, or hickory nuts; nor any other American food staple had found their way to the Old World; even the beeches, chestnuts, oaks, and maples were distinct; and the same is true of the New World ground nuts and the grapes, which were the parent species of our delicious American varieties. Quite unlike anything in the Old World were such cultivated plants as the Cactaceae, the capsicum peppers, and the manioc from which cassava is made.

In Polynesia the evidence thus offered by cultivated plants points to the spread of Asiatic culture eastward across the Pacific, while the peculiarities of the cultivated plants of America point to its isolation from all the rest of the world; an isolation which is further established by a radical dissimilarity of all American languages from Old World linguistic stocks. In no language of the New World, for example, is there a vestige of Hebrew, which would support the cherished theory of the migration to this continent of the lost tribes of Israel; nor is there a suggestion of any linguistic element to indicate connection with the Chinese, nor any relationship between the builders of the American pyramids and those of Egypt.

There are many distinct groups of American languages. Very often the language of a tribe is quite unlike that of its nearest neighbors; while at the same time it may resemble the languages of tribes quite remote. This fact indicates former segregation of the various groups speaking the unlike languages and a common ancestry or close association of the tribes speaking the allied dialects. As examples, I might mention the Quichua Indians of Peru, whose language is very unlike the languages spoken by the Arawak and Carib Indians to their northward and, at the same time, quite distinct from the languages of their Brazilian neighbors to the eastward. The Aztecs of Mexico spoke a language differing radically in structure as well as in vocabulary from the Maya language of their Yucatan neighbors; yet there is unquestionably a relationship between the Aztecs and a number of very distant tribes, shown by resemblances of their languages, as in the case of the Shoshone Indians of the northern United States and the Nuhuatl tribes of Salvador and Costa Rica. In the same way, the Algonquian dialects, which differ greatly from those of the Iroquoian, show a close relationship between very widely scattered tribes in North America, from

North Carolina to Quebec. Such resemblances and radical differences point to a very remote and long-continued segregation which permitted the independent formation of distinct linguistic stocks; while the antiquity of man in America, both north and south of the equator, is further attested by the development of such a cultivated and highly specialized food staple as maize, whose ancestral prototype we have sought in vain. Its endless varieties, fitted for widely diverse conditions of soil and climate, also point to a long period of cultivation in dissimilar culture-areas, which enabled them to adapt themselves to conditions very different from those of the original stock from which they sprang.

All this evidence points to the peopling of this continent at a very remote time, perhaps as far back as the close of the Glacial Epoch; and it also indicates that the early progenitors of our Indian tribes had left their original homes in the Old World before any of the linguistic Old-World stocks had taken shape; before Sanscrit was Sanscrit; before the languages of China or any other Asiatic people had become established; and just as in this hemisphere the natives developed their own languages from the most primitive elements of speech, so most certainly did they develop their agriculture from the wild plants of the fields, the swamps, the hillsides, and the forests. In both respects, as I have already pointed out, they differed from the Polynesians who brought with them to their island homes not only their language but their agriculture, from the cradle of their race in the Malay Archipelago; cuttings of seedless breadfruit and of sugarcane, fleshy roots of taro and yams; even trees, like the Indian almond and the candlenut.

Here I would like to point out to the members of the Nut Growers' Association the chief difference between nuts and other food staples. Nearly all of our cultivated vegetables, including maize, beans, potatoes, sweet potatoes, squashes and pumpkins, are annuals, sensitive to frost, which must be raised from seed each year, and which differ so greatly from the primitive plants from which they came that their ancestral forms cannot be definitely determined. Most of these vegetables are in all probability of hybrid origin, the result of cross pollination and selection. In the case of our native nuts the conditions are quite different. We know the original ancestor of the pecan, our hickories and our walnuts. The fine varieties now cultivated are not hybrids but have been selected from wild trees. In connection with nuts I would also point out that in all probability they were the most important food-staple of primitive man, as well as of his simian ancestors. It required no great intelligence to gather them or to store them after the fashion followed by squirrels. Intelligence, however, is required to plant nuts and to transplant nut trees. Still greater intelligence is involved in the process of preparing certain nuts for food. A delicious creamy emulsion, for instance, was prepared by the Virginian Indians from hickory nuts. Cracking them and removing the kernels was too long and tedious an operation; so they developed a method of gathering them in quantities and crushing them in a hollowed log, together with water, pounding them to a paste and then straining out the fragments of shells through a basket sieve. The milky fluid which was thus formed was allowed to stand until the thick creamy substance separated from the water. The water was then poured off, and the delicious cream which remained was used as a component of various dishes. This substance was called by the Virginian Algonkian Indians "*Pawcohiccora*," a word which has been abbreviated and modified to "*Hickory*," the name by which we now designate not only the nuts, but the tree and its wood.

It is interesting to note that a similar creamy or butter-like substance was derived by a similar process from various palm nuts in Central and South America. Cieza de Leon describes such a process in his Chronicle of Peru, in connection with a nut which was described as *Cocos butyraceæ*, but which was not a true *Cocos*, or coconut. Long before the discovery of

America, a somewhat similar process was used in the Nicobar Islands for extracting a creamy substance from the grated kernel of the true coconut, *Cocos nucifera*, which in early times was called *Nux indica*. This process is still followed throughout Polynesia. Some of the most savory dishes of the Samoans and the natives of Guam are enriched and flavored with this coconut cream, which is a substance quite distinct from the water, or so-called milk, contained in the hollow kernel of the nut, which is so commonly used for drinking.

Coming back to America, I would call attention to the value of some of our native pine nuts and acorns as food staples. Certain Indian tribes of the Southwest live upon pine nuts at certain seasons when they are ripe. Dr. C. Hart Merriam has told of the utilization of acorns by various tribes of Indians in a beautifully illustrated article published in the National Geographic Magazine, 1918, entitled "The Acorn, a Possibly Neglected Source of Food." "To the native Indians of California," he says, "the acorn is, and always has been, the staff of life, furnishing the material for their daily mush and bread." He describes the process of gathering and storing them, shelling, drying, grinding the kernels, leaching out the bitter tannic acid, and preparing the acorn meal in various ways for food. In eastern North America, several species of acorns were somewhat similarly used, including those of the live oaks of our southern states. The Spaniards of Florida sometimes toasted them and used them as a substitute for chocolate or coffee. Chinkapins were used for food by the earliest English colonists. They are mentioned by Herriot, the historian of Sir Walter Raleigh's colony at Roanoke. In addition to these, the early colonists learned to eat the so-called "water-chinkapins", which are fruits of the beautiful golden-flowered American lotus, *Nelumbo lutea*, a plant closely allied to the sacred lotus of India, China and Japan, whose nuts are even now used as a food staple. The split kernels of the latter may be bought in the Chinese shops on Pennsylvania Avenue in this city. The rootstocks of both the American and the Oriental lotus are also used for food. They resemble bananas joined together end to end, with several hollow longitudinal tubes running through them.

Before I close, I should like to call attention to a plant, endemic in eastern North America, whose tubers were called "ground-nuts," or "Indian potatoes" by the early colonists. The latter name caused the plant to be mistaken by certain early writers for the white potato, which was unknown in North America in early colonial days, but which was confused with the ground nut on account of the resemblance of the descriptions of the two plants. The white potato, *Solanum tuberosum*, was discovered in the Andes of South America by Cieza de Leon; it was quite unknown in North America or in the West Indies in the days of Sir Walter Raleigh and Sir Francis Drake, both of whom have erroneously been given the credit of introducing the potato into England. The "potato" which they observed in the West Indies was not *Solanum tuberosum*, which we now call the "white potato" or "Irish potato," but a very distinct plant, *Ipomoea batatas*, which we now call the "sweet potato," but which in early days was known as the *batata* or *potato*. The error which has become widely spread, can be traced to John Gerarde, the first author to publish an illustration of *Solanum tuberosum*. In his celebrated *Herball* he declares that the potatoes figured by him were grown in his garden from tubers which came from "Virginia, or Norembega." It is quite certain that this statement was untrue, and that, as certain English writers have already suggested, Gerard "wished to mystify his readers." Whatever may have been his motive, the error became widely spread. Even Thomas Jefferson was led to believe that *Solanum tuberosum* was encountered in Virginia by the early colonists, and Schoolcraft declared that its tubers were gathered wild in the woods like other wild roots. The Indian potato of the early colonists is still abundant in "moist and marish grounds," as described by Herriot. It is a tuber-bearing plant of the bean family, and is known botanically as *Glycine apios*.

But I fear my talk has become too discursive, in turning from nuts to ground nuts, and from ground nuts to potatoes; but the subject, bearing as it does on the origin and history of cultivated plants, is one which has great attraction for me, and I hope it may have been of interest to the members of this association.

Professor C. P. Close, Pomologist, U. S. Dept. of Agriculture, spoke as follows:

MR. CLOSE: The subject I had intended to speak on was "Extension Work in Nut Growing." Many of you know that I am putting in most of my time on the fruit end of extension work, but I am also doing some extension nut work. I was hoping that there would be representatives from many of the states here, because I wanted to encourage them to get in touch with the state extension men, to work up interest in nut culture.

My talk will be very brief, but I would like to mention that very few of the states as yet are doing extension work with nuts, especially in the North. Some work is being done with pecans in the South.

I have been astounded in talking with the landscape men in the North to find that they have not considered nut trees as ornamental trees. But after I mentioned that a walnut or a hickory or a pecan tree is an ornamental tree, and just as much so as the elm, the oak, or the maple, they thought it would be a good idea to use them and agreed to recommend the use of nut trees as shade, lawn and roadside trees. Then I suggested the filbert for clump planting as an ornamental. I hope in the future that nut trees and filberts will be used more extensively by the landscape extension men in their work throughout the country.

In most of the states there are fruit extension specialists but only an occasional landscape extension specialist; so I try to interest the fruit men in the planting of nut trees, and a few of them are doing this, particularly in Indiana, where the fruit extension specialist has been interested in having pecan and English walnut trees planted in school yards. It seems difficult to get people to comprehend and practice nut tree growing and to understand the various uses of nut trees. We can judge from the small audience at this meeting that there are not enough people interested in nut growing. In my journey throughout the country I occasionally run across men interested in growing a few nut trees, and I try to induce them to become members of this association; but it seems to be a hard thing to do.

A few days ago I called on a man in New Jersey who said he would have twenty bushels of hickory nuts and two or three bushels of English walnuts if the squirrels did not take them. He is up against a state law which protects the squirrels but does not protect him.

I wish we could send out word with you to the states to get at least a few people interested in nut culture, and have them write to the agricultural colleges and the experiment stations and arouse some interest along this line at those institutions, not only among the fruit extension men and the teachers, but also among the landscape men as well. There ought to be more interest taken in this work at our colleges and universities, and nut culture courses ought to be organized. The foresters ought to be induced to use nut trees wherever possible.

That is all of the time I care to take at present, Mr. President, but I wish to say that if there is any way of arousing interest in the states, I would be glad to carry the word from Washington and to push it just as hard as possible.

Hon. W. S. Linton, Saginaw, Michigan, spoke on "Roadside Planting vs. Reforestation," as follows:

As a delegate to the National Tax Association convention at White Sulphur Springs, it has been my lot to have been named on both federal and state committees, with the idea of exempting from taxation those who would produce trees for the future. My experience has been that exemption from taxation for the purpose of producing our future forests is a wrong one. The sentiment of the people is against exemption from taxation, and I do not know how it may be practically applied to the growing of the forests that our country must have in the future. But the individual will not carry out the work, and the corporations will not undertake it, so it devolves upon the government of the state to reproduce those forests. The government lives for a long period in between many life-times, and ours should live as long as the earth. It is therefore up to us to reproduce those forests which we once had and, as all things come back to the state, then the state should reforest.

Next the roadways are to be considered. Roadways will grow a better class of timber and trees; they are rich in soil, generally, because they pass through the most fertile regions of the country and, up to this time, they have been waste land. I believe that the farmer is right in his wish that trees which shut in the roadsides should be cut away, that the sunlight should be let in and the roads hard-surfaced. We saw in our trip that where the trees shaded the roads they were almost impassable at times, while in the open places, they were fine.

In Michigan we took up the question of roadside planting, and Senator Penny fathered the bill, the pioneer measure, that caused our state to plant roadways. We have a very competent landscape engineer in charge of one of the departments, and he is planning to grow roadside trees, using nut-bearing trees, so that the next generation will profit largely by the work of today. And this is just because of this association.

When I was honored with your presidency, one of the features of the work we carried on was in getting nut trees from historic places, especially from Mt. Vernon. The Superintendent of Mt. Vernon very kindly told us that we could have the walnut crop from trees that were started there during Washington's time, and the only stipulation was that we should not commercialize the idea; that those nuts were priceless, and that we should not receive any money for them, but should distribute them in the schools and in a public way cause interest in the planting of nut trees. That very movement brought about wonderful results, and today there are from five to ten thousand walnut trees growing in our state, about the height of a man, all of them having come from Mt. Vernon.

On our way through from White Sulphur Springs, we passed through the home of Thomas Jefferson, Monticello, and we found some magnificent nut trees planted by Jefferson. Some of our best trees today are from those given to Washington by Thomas Jefferson; and I arranged at Mt. Vernon to secure some of the nuts from the trees Jefferson planted there.

Just yesterday Mr. Dodge, the superintendent at Mt. Vernon, again said that we could have the crop for this year. We will have a number of bushels from there, although the trees have not been as fruitful this year as usual, and I leave it to you to judge as to what we should do with those nuts this year. Some of you have ideas about this, and I would be glad to adopt them. But when the fact is known that the walnuts can be secured in that way the entire country will want them. At present I have letters from Texas and other places asking for some of Mt. Vernon's nuts. It is a movement that will cause more people, in my opinion, to have nut trees than any other, and we should push it to the limit.

I had a letter from Henry Ford's secretary, asking for a dozen trees which might be planted at Mr. Ford's place in Michigan. Mr. Ford is doing great good, so far as the saving of the forests is concerned. He has immense tracts of land where he is caring for every root and branch.

Letter from C. F. Bobler, Landscape Engineer in Michigan:

The laws of Michigan, as you are well aware, encourage the planting of trees and shrubs by the highway authorities, and protect existing roadside trees from injury or destruction. Under those laws considerable planting has already been done, and in such planting a liberal use has been made of the nut-bearing varieties of trees, especially the black walnut, which is indigenous to much of Michigan.

Besides the economic value of nut trees, on account of their food products while growing and their timber products when mature, they are generally very attractive in appearance, and, therefore, very well adapted to roadside planting.

Roadside development presents a field for considerable study to produce plantings which afford a variety of effects in trees and shrubs, by using varieties best adapted to the soil and climatic conditions, which best harmonize with the local topography and which to a considerable extent have an economic value in addition to their ornamental value. Nut trees admirably fulfill these requirements for roadside planting and while I believe that such other desirable varieties of trees as the American elm, the sugar maple, and others, should be used in proper proportions, I am fully convinced that the varieties of nut trees adapted to our soil and climate should be used liberally in the planting of the roadsides of Michigan.

The plans for the future development of the state trunk line highways in this state, contemplate the planting of the black walnut, butternut, sweet chestnut, hickory, beech, and other varieties of nut bearing trees in considerable quantities, and I am confident that their use will add to man's enjoyment of the highways and that these trees will become an economic asset to the regions where they are planted.

THE PRESIDENT: There is one thing Mr. Linton mentioned that I wish to put special emphasis upon; the distribution of trees grown from Washington's home. Last year Mr. Jones sent out a lot of seedling walnuts and there are quite a few in Rochester. It was delightful to see the interest manifested by the people receiving those seedlings and to hear how the people were succeeding. Some of them have written me.

MR. REED: Possibly it would help if, when any of us here present should chance to visit historic spots, we would get nuts from such places and send them to Mr. Linton; from Gettysburg or any of those places. We should each consider ourselves committees of one to get those nuts and to deliver them to Mr. Linton.

MR. BIXBY: I will see what I can do about it, and will get some of the nuts today.

MR. O'CONNOR: I do not know how Mr. Linton would feel about sending to different schools some of the nuts that were given him by the superintendent at Monticello, and in letting the children have a little nursery, and the means to beautify their home towns, but I will say that if you get the children started in a thing like this, you will have the parents following up.

MR. LINTON: There is another point I wish to mention. Mr. Dodge sent one bushel of the walnuts which he said were taken from a particular tree that he admired. He thought it was the best variety of all of them. That tree, a year ago, was struck by lightning; so he requests that some of the trees produced from the nuts of that particular tree, be sent back to Mt. Vernon, in order that he may have some seedlings from the original tree. It is a fact that those nuts produced the best yields of any that we planted in Michigan, showing that the seeds from the best tree will bring the best results.

ENCOURAGEMENT FROM FAILURES IN GRAFTING

Dr. G. A. Zimmerman, Piketown, Pa.

After improving from an illness of several years, and feeling tired, impatient and at times discouraged with progress in my physical condition, last spring I secured a few bunches of scion wood and turned to my old boyhood hobby for diversion; this time, however, by working on nut trees instead of fruit. In presenting the following at the request of others, I do not claim any originality, but simply draw the attention of interested parties to some possibilities and probabilities. My results have been very variable and many of them show as successful a failure as any one could possibly obtain. The scions referred to in the following tabulated record were put in from May 20th to July 20th and were well "mixed together" in the hope of giving better opportunity for cross pollenization, a few of every variety except the Hales being put in every day. The Hales were all put in late in July. I have grafted many other varieties of fruits and nuts but a record of the hickory only is shown below:

	No.	Growing	Died	% Growing	% Died	
Weiker	46	0	46	0	100	One graft to tree
	5	3	2	60	40	T.W.T 1-1/4" diameter
	5	1	4	20	80	U.W.T.

23 1 22 4.2 95.8 U.W.T.
Taylor 5 2 3 40 60 U.W.T. 10" diameter
27 7 20 25.9 74.1
Fairbanks 15 11 4 73.3 26.7
Vest 27 1 26 3.7 96.3
Manahan 22 7 15 31.8 68.2
7 0 7 0 100 U.W.T. 3" diameter
Laney 13 6 7 46.1 53.9
15 1 14 6.6 93.4 U.W.T. 6" diameter
Beaver 5 2 3 40 60 Scions poor. But one
grew 7 ft. 4 in.
Kentucky 19 7 12 36.8 67.2
10 1 9 10 90 U.W.T. 5" diameter
Kirtland 12 5 7 41.6 58.4
16 5 11 31.3 68.7 U.W.T. 5" diameter
7 1 6 14.2 85.8 U.W.T. Put on late
as also the Hales
Hales a 6 1 5 16.6 83.4 U.W.T. 3" diameter
b 35 0 35 0 100 U.W.T. 10" diameter
c 2 2 0 100 0 T.W.T. 1-1/2%" diameter
d 4 4 0 100 0 T.W.T. 2" diameter
e 3 3 0 100 0 T.W.T.
f 3 2 1 66.6 33.3 T.W.T.
g 6 4 2 66.6 33.3 T.W.T.
—— — —- ——- ——-
Total 338 75 263 22.2 77.8

The last two series of the Hales made 100% start also but bugs killed three grafts.

U. W. T. means a tree from which all the lower limbs were cut back to about a foot or eighteen inches and grafted, a few top limbs having been left intact.

T. W. T. means a tree from which the top had been cut, the lower limbs and stub having been grafted, although a few of the lower limbs were not sawed off.

A study of the above record is interesting. All of my stocks are of the mockernut type, varying from three-fourths to two inches in diameter, except a few trees to which I refer specially as T.W.T. and U.W.T. It will be noted that the Weiker and the Vest made the poorest catches. It could not have been due entirely to weather conditions or the condition of the scions, for the scions of these two varieties were equal to anything I had. In view of the fact that they are both very desirable nuts, I always carried a few scions and kept placing them frequently as I placed other varieties. Many Vests were placed at the same time as the Fairbanks, which shows 73.3% catches. The one Vest that did catch, however, made a very thrifty growth, showing that it is possible apparently to do well on the mockernut.

With the Weiker, about the 15th of July, I put five scions on the limbs and trunk of a tree about 1-1/4 inches in diameter, the top having been cut out, with three catches, 60%, against another lot of 46 with 100% failure and 23 more with 4.2% success. Such antics are difficult to understand.

Many of the scions were put in the trunks of the trees; others were put on the small branches with the splice graft. The scions placed on the trunks, or the larger limbs near the trunk, apparently did somewhat better than the splice grafts further out on the limbs. In the walnut and other sappy trees, however, the splice graft out on the small limbs did better.

It is of peculiar interest that all of the large trees from which the lower limbs were sawed and the stubs grafted, the topmost limbs having been left, designated as U.W.T., did badly. While in the case of the five Hales, three had 100% and two had 66.6% catches. These two also had 100% catches but bugs ate the tender shoots and killed three of them. These trees had the tops cut off last fall leaving only a few lower limbs. They were put in on July 20th after the sprouts had well started on the trees. The sprouts were not taken off but their tops were pinched out. These grafts made a growth of from one to two feet or more. At the same time a tree was trimmed (Hales b in the record) and all the lower limbs grafted with Hales, leaving a few top branches only. Thirty-five were set and not a single one grew. The location of this tree was better than any of the five above referred to, because a couple of those trees were standing on the top of a rock where one would wonder how they could exist, and it was so hot when I placed the grafts that I had to quit and get out of the sun. In spite of that 100% grew.

A study of the above record leads to the conclusion that there is very little difference in plant and animal cells and it seems clear that certain old, underlying principles must be dealt with. I need not refer to heredity because, while it is undoubtedly quite possible, perhaps, to influence heredity tendencies so as to get stocks to accept scions more readily, it is not the major issue for most of us just now. Next spring we will take what heredity has given us and be satisfied. However, it appears certain that our results in grafting the various stocks we now have will depend largely on our ability to:

> 1. Regulate plant circulation. 2. Stimulate cellular activity to a point compatible with wound repair, defensive and growing processes. 3. Control plant cell nutrition.

One of the very first things we physicians do upon seeing a patient is to investigate his circulation. If the pressure is too low or too high, for any reason, we immediately take measures to correct it, because we know that disastrous results will quickly follow if that is not looked after. Plant circulation, or sap flow, is no less important. Mr. Riehl, Mr. Jones and Dr. Morris made great strides when they advanced the ideas of covering the wound and the scion completely to prevent evaporation, thereby also controlling the sap pressure. With the exception of shading, pruning and defoliating, this is about the only method we have of preventing evaporation. Defoliation, of course, interferes with the tree's power of growth. Controlling the humidity is probably not practical on a large scale.

A proper and careful cutting of the tree beforehand is important. It appears that to cut the top completely out while the tree is dormant, so disrupts the routine circulation that the few lower branches which are left intact, are well taken care of and, it seems to me, that this, together with the stimulation of WOUND REPAIR by cutting and allowing time enough for the cells to get into action, was the prime reason for the 100% success in the three Hales and the cause of the 100% failure in the other Hales tree.

Other methods of controlling the circulation are of course drainage, irrigation, mulching, location of the orchard, placing of condensers of moisture, such as stones and other hard substances beneath the trees, and many other contrivances which are in use, and which I shall not discuss.

With reference to stimulation of cellular activity we are considerably concerned. In medicine I have found the subject of wound repair and immunity most interesting, the two subjects seeming to be more or less related. Some animals will repair wounds and immunize readily, while others will not. In a general way young healthy animals and human beings immunize most readily, while older ones frequently fail almost entirely.

Interestingly enough plants seem to be strangely similar in this respect, and the thing that stimulates cellular activity for defensive purposes (immunity) apparently stimulates growth and wound repair. The thing that stimulates most actively for a special purpose is the thing itself, the best stimulant for wound repair being the simple injury. To illustrate briefly: In my work last summer I came in contact with two enemies, yellow jackets and copperheads. The copperhead stimulated me to carry a club in defense, while for the yellow jacket the club was of little value and I rather preferred carbon bisulphide. Had I ignored my senses and allowed nature full sway, as a tree does, the snake would have injected his venom and the yellow jacket his toxin, and my cells would have accepted their only alternative and proceeded at once to build up a specific defense, after which they would have been in better shape for development, providing the poison would not have been so great as to prove fatal. Injury to a tree certainly does stimulate wound repair, defense and growth. It is well known that trees with many transplantings, root injuries, transplant much more readily, and the nurserymen use this method of stimulation as a routine procedure. I learn in Florida that in order to transplant a good size palmetto, they are in the habit of digging down on one side and cutting the roots the year before removal. It will then transplant more readily. Pruning has the same cell stimulating effect if done at a time that will retain the stored nutrition. An attack of disease just as surely stimulates cellular activity and growth but it is too frequently followed by disaster.

We have all heard of driving rusty nails into trees (thinking the iron produced the beneficial results), cutting a slit in the bark of the limbs and trunk for "bark bound" so called, etc., all of which have stimulating effects with more or less permanent injury to the tree. Who knows but what the sap sucker, with his ability to dig into the bark and extract a piece of cambium, was not sent to us to aid in preserving our trees by stimulating new growth?

In my work last summer trees that were subjected to slight injury before hand apparently accepted a larger proportion of grafts. I will briefly cite two specific illustrations. A little butternut tree located near the house was the object of my efforts for over two years. During my illness I frequently went out and pruned a few branches or put on a few buds. Something would happen to me and possibly I would not see it again for months, and in the meantime the buds would be strangled or knocked off. Another little hickory tree stood in the roadway. Harrows, plows, wagons and even logs were dragged over it. Grafts on both these trees caught rather readily last spring. In fact two black walnut grafts on this little butternut were two of the very few that I got to grow at all last year. My walnut grafting was almost a total failure. I have this to say, however, that I had no dormant walnut scions, my scions all being cut in May or June.

Mr. Jones, by marking the site of his patch bud several days in advance, admirably carries out this idea by locally stimulating the cambium cells. Dr. Morris's scheme of using white wax, besides regulating sap pressure, allows the actinic rays of the sun to stimulate cellular activity. Cutting the top out of the tree, which disrupts the normal circulation and throws it into the few lower limbs, besides stimulating the cells into activity, has apparently in a large measure accounted for the slight success that I have had. Other methods such as injecting some substance under the bark, applying antiseptics, or some stimulating chemical in a similar way, as "Scarlet Red" is used in skin grafting to increase epithelial growth, may aid materially. Certain chemicals applied to the tree and leaves, as used in sprays, seems sometimes to stimulate growth in a way that can hardly always be accounted for by the checking of the disease for which it was placed.

Much more could be written on cellular stimulation but enough has been said to encourage others to make observation in this connection, for it is highly probable that the lack of proper stimulation of the cambium accounts for more failures in top working trees than we are aware of.

3RD CONTROL OF PLANT CELL NUTRITION

With this topic we are probably less concerned in its relation to grafting than when the growing and bearing stages come. However, certain nutritional disturbances appear early and the more vigorously the stock is growing beforehand the better progress, of course, the grafts will make when they are started. Whether or not they will start more readily have I been unable to ascertain, but I have a bunch of little fellows with a growth of only an inch or so, and so puny that I cannot account for it in any other way than a lack of proper nutrition. Many of these little trees, used as stock, are very old in comparison with their size and they will probably be dwarfs all their lives. It is a question whether many such trees should be grafted at all. Further observations will have to be made to decide that point. Perhaps proper preparation for a year or two would be beneficial.

This topic will largely be left for future discussion under another subject, but it occurs to me that much might be accomplished by proper attention to nutrition, especially when setting out trees for grafting, selection of proper site, fertility of soil, cultivation to aid absorption, etc. I have observed limbs of animals much smaller than normal due to prohibited movements or lack of proper circulation, one side of a tree developed out of proportion, eggs without hard shell due to lack of calcium in the hen's diet, and I know of an old English walnut tree that bears nuts with shells so thin as to be almost negligible. I am told that at one time this tree bore a nut with a much thicker shell. It has never had any attention and it is quite probable that the lack of proper shell building elements causes the trouble. I have grafted a few of these and I want to see what happens by furnishing better nutrition.

Concerning scion wood, I have "ringed" some limbs, similar to the method used sometimes in producing extra large fruit, in an effort to have the scion store up a large amount of nutrition. This experiment I shall continue in the spring.

This article is based entirely on my own ideas, observations and conclusions in connection with old standing principles. As previously stated, I claim nothing new and my only desire is to stimulate others to make like observations.

Carrying out my conclusions in my work next spring I propose to cut the tops out of all my trees, leaving a few lower limbs instead of the top ones, allow them to start growth a little before grafting, pinch the tip from that growth, and, in addition to covering with paraffin or some combination of it, shade the scions on the south-west side, either by tipping branches over them or some other way. Paper bags seem to absorb the paraffin. Double grafting in the case of the Vest and the Weiker will be tried. Whitewashing the stock to prevent sun burn will be used where necessary. Several other experiments based on the idea of cellular stimulation before the scions are placed in position will be tried.

Dr. M. B. Waite, of the Federal Insecticide and Fungicide Board, U. S. Department of Agriculture, spoke as follows:

DR. WAITE: Some of you may recall that several years ago, when you were meeting here in this hall, I gave you a paper on the nut diseases of the northeastern part of the United States, and it would not be desirable to go over that same ground again. At that time, we took up the bacteriosis of the

Persian Walnut, and filbert blight, and I outlined a program of proposed treatment for the filbert blight. It might be interesting to note here that Dr. Morris, and I believe also Mr. Bean, put that treatment into practice with success. The situation still remains, however, that we do not know of diseased plantings of any size. If we find a real plantation of filberts we will be glad to attempt control measures ourselves. I have planted about two dozen filberts and they still remain free from the disease. There are very few local hazel nuts, wild or cultivated, around Washington; but we understand that the few hazel nuts are free from this disease.

There are two or three things I wish to mention. One is the repeated inquiries reaching my office with regard to the non-filling of nuts, mostly the cultivated nuts, sometimes the pecan, sometimes the black walnut, and frequently the English walnut. The subject is a complicated one and the disease is not one that we can put under the microscope and diagnose at once. The trouble is due to a complex of varietal and environmental conditions, the effect of the conditions of growth, of soil fertility, temperature, soil, water and humidity, sunshine, etc., on that plant. Very often it is because people get the wrong variety and do not know what they have. They may have an unproductive seedling.

On the other hand a good variety may fail to bear in a locality where it is not suited. Very frequently the real lack is in soil fertility. Of course the success of the pecan trees down South around pig pens is an old joke to you gentlemen, but there is truth in that. For good nuts there is often need for a little extra manure or fertilizer, or perhaps both. Sometimes there are rich pockets in the earth where those trees would like to grow, or rich bottom lands which will produce without manure. I think one of the best ways is to fertilize with manure, if possible. Pollination troubles in connection with the non-filling and dropping of the nuts should be thought of.

Then there is another angle to be considered, and perhaps I can express it most definitely to you by citing the example of the June drop of peaches. Whenever a tree, like the peach tree or the pecan or the black walnut, sets its fruit in the spring, you will find that there are cross-pollinated and self-pollinated fruits. These will begin to drop their nuts or their fruit at definite stages. Furthermore we will find the abortive seeds are not one size. This means that there were definite stages of the pollination and of the fertilization. I should like to work that up and find what the stages are.

The last big step in the dropping of the peach tree is the shedding of the fruit just as the pits are hardening. When they are hard the fruit does not fall. So this June-drop question ties in with the complications of pollination and nutrition. We know from experiments on the sterility of the pear tree, if highly fed and cultivated, such as those I worked on in the city of Rochester, that those highly fed trees will have some self-fertilized pears. In all of the pears we got no pears resulted when pollinized with the pollen of the same variety, except on those well fed trees. We learned this in the East, and have since found the same type of self-fertilized pear occurring naturally in California and other places in the West. In nut production that whole question of setting and filling is tied up in a complicated way with pollination and nutrition.

Aside from nutrition the other thing to be considered is that of disease. The common black walnut around Washington is generally poor from fungus leaf diseases. Those of us familiar with it around here know that they do not fruit well. This is not a good place for the common black walnut. The wild ones are nearly all poor. I was raised in the Mississippi Valley, where there were large nuts and fine ones, and we gathered those which fell from the specially good trees. They do not grow so well here, except the Stabler and a few others.

Leaving that subject, there is another I wish to take up. That is, the great number of complaints about winter-killing of the English walnut. Wherever we have been able to trace that down, as we frequently have, we find that the English walnut suffers more from winter-killing right around Washington, D. C., and in Pennsylvania, than up in Rochester; and we also have complaints of winter-killing as far south as Georgia. A common cause is the variation of moisture. After a dry spring and early summer soaking rains come in August and September, and the trees, brought suddenly into growth at the close of the season, when they should be drying out, the walnut tree in particular, show winter-killing. So I think one of the main troubles with the English walnut in the Eastern United States is the winter-killing. Even in Georgia we may have this trouble with the pecan, young trees two and three years old, and I have photographed them.

As to false stimulation, in the woods, where these trees grow native and under the conditions to which they are necessarily adapted, they are mulched and crowded when young by their competitors. In cultivation we do not get the crowding and the mulching that makes steady growth and proper ripening. So you should, by some process, growing corn, cover crops, or other trees, keep your delicate nut trees a little crowded and, if possible, mulched while young; and then later, cut out the undesirable things and let the trees have room.

I am not fully prepared to speak about the nut work of the Bureau of Plant Industry, because that should be handled by the chief of the bureau. I have charge only of the diseases of fruits and nuts. We have had $8,200 allotted to the project and will have $2,000 more this year, making $10,200. Originally that was $3,000 for nut diseases all over the United States. We started to work mainly on the southern pecan diseases, and partly on the bacteriosis of the walnuts of the United States. But the Southern Pecan Growers' Association got some additional money for the bureau, $5,000 of which was given to the fruit disease investigations, and was tied up with the other $3,000. But the wording of the bill said, "All for pecan diseases." So we transferred more to the project and made it $8,200 for the nut diseases. That means we have done very little work for the nut diseases except on Southern pecans, and I have been warned that one must not stress southern pecans with the Northern Nut Growers' Association.

We have had, however, one man, and will have two men, on the southern pecan diseases in Georgia, on pecan scab and pecan leaf diseases, who are winning out beautifully, and have nearly solved many of the problems, including the pecan scab. One of the difficulties is the occasional late summer rainy spell, bringing diseases and bad conditions. But in general we have solved the problem pretty well.

Then we have the more permanently dangerous disease, pecan rosette, which has taken about half of the pecans in some sections of the South, especially in south Georgia and in Florida. That disease is being experimented upon in the most extensive way of any of our projects. There is only one word to say about pecan rosette, and that is—humus—the disease is cured by the application of humus.

MR. REED: How far north is the walnut rosette disease?

DR. WAITE: As far as Falls Church, Va., but not much in the North.

MR. REED: The question was asked yesterday as to whether it could not be overcome in this latitude.

DR. WAITE: That nobody knows. The soils east and south of Washington are all acid, and the conditions are wrong for rosette. The soils have no tendency to chlorosis. They are, in fact, antichlorotic. Theoretically you could get the rosette conditions in the Piedmont region, but you are almost certain not to find them over this way.

Now in the organization of the Bureau of Plant Industry there are at least two main offices where nut problems would be studied; in the Division of Horticultural and Pomological Investigations and in my office, where the diseases are studied. Remember, also, that the insect pests are studied in the Bureau of Entomology; they have experimented quite extensively with pecan insect pests, and have the organization to handle such pests. Of course there is a Bureau of Markets and the Office of Soil Fertility in the Bureau of Plant Industry, which handle the pecan, incidental to the other studies.

MR. BIXBY: I would like to ask Dr. Waite a question. The association has spent a good deal of time in developing exact methods of measuring quantitatively the various characteristics of nuts which are considered valuable, and that study has given us methods of comparing notes from year to year, comparing the same nut, and I have noticed that it is quite frequent that the kind of nut that is good one year, will not be so good the next year. To take an example, the Clark hickory, which took the prize one year, the next year fell so far down that it would not take any prize. But after a good deal of trouble I found that by careful examination I could pick out from the nuts a few which tested up as they did before. It occurred to me that a condition of that kind would be more likely to be due to difference in the soil than in the fertility of the pollen. Dr. Waite has had more or less experience in noting the effect of the pollen, and I would like to ask if he thought this the cause of the difference in the nuts.

DR. WAITE: I think it might be the cause for a little difference, but we could account for the difference by entirely different things. By environment and other conditions. Take the apples grown in this vicinity; I have observed that certain seasons fit certain varieties. This year it was favorable for Ben Davis, and yet we have had a poor crop of most varieties; the conditions were bad for the Winesap to set, but yet the fruit is good. Every year and every day is different; and plants are subjected to these complications, and the yield, or the result in fruit, is a response to environment. They are so very susceptible to these things. I came here this morning after picking some cross pollenated pears on the Arlington Farm. We have a lot of crosses there where we study the hybrid seedlings. Some will be almost too poor, in certain years, to deserve further attention, and good another season. In other words, these nuts probably do not vary any more from year to year than many of our fruits and vegetables do, and the main factor is probably response to environment, namely, temperature, air humidity, soil moisture and sunshine.

THE PRESIDENT: I might mention that we have had a filbert orchard at Rochester for eleven years, and there has not been the slightest indication of blight there yet.

MR. REED: I would like to ask Senator Penny how the Roadside Bill is taken in Michigan.

SENATOR PENNY: According to the Michigan law, the people along the roadside consider that their property is subject to the right of transportation on the highway; just as a stream is owned by individuals in Michigan, subject to the right of individuals to use it. This bill says, "Give the right to plant trees on the highway," and I think the planting is done with the consent of the owner. The agricultural college has a landscape gardener connected with the landscape department; he will have charge of planting along the roadside, and I think it will be done in a scientific manner; but I believe it is necessary to get the consent of the owners first.

MR. BIXBY: Last evening Mr. Franklin Weims, of Washington, was with me on the state highway of Maryland, coming south from Baltimore. The highway is being constructed at the rate of about eight miles a year, and funds have been provided. Mr. Weims feels that something should be done to see that the new highway is properly planted with trees, preferably nut-bearing trees. I was thinking that the association might, by some resolution, bring that matter to the attention of proper authorities. I would like suggestions.

MR. CLOSE: It might not be out of order to adopt a resolution and address it to the Governor of the state, Governor Richie; and also to the State Forester, Dr. Besly, suggesting that perhaps some of the trees and seedlings might be presented to the state, some of the trees that Professor Linton spoke of this morning. Trees of that sort might carry some weight.

THE PRESIDENT: Suppose we adopt a resolution and name Professor Close to take up this matter with the proper state authorities, speaking particularly of our ability to furnish seedlings from the Mt. Vernon trees.

MR. CLOSE: If it is the wish of the association, I would be glad to do that. (Motion made, seconded and adopted).

LETTER FROM F. H. WIELANDY, ST. LOUIS

Gentlemen:

First of all I congratulate you most heartily on being members of an organization which means so much to the public, as consumption of nuts is largely increasing and I much fear that the present day production is not in line with the demand.

Although only a nut culturist by proxy I have manifested a deep interest in this for many years, which is exemplified by the fact that on my different hunting trips, in which I have indulged for over thirty-five years, in the past twenty-five years I have also made it a point in the fall of the year, to have with me a large pocket full of such nuts as I thought would more easily come up and benefit some one in the future. I usually carried with me black walnuts, hickory nuts, pecans and acorns, and in my rambles through the woods and along the highways, I would plant these where I thought there would be less chance of their being molested if they developed.

In going over the same ground quail shooting, last fall, ground that I had covered more or less for a good many years, I began to see the fruit of my efforts, and felt repaid many fold for what I had accomplished.

Unfortunately we are a nation of destruction, rather than of construction, so far as our timber is concerned, and this is more noticeable in fruit and nut trees than in other varieties; although, being interested chiefly in these I possibly am biased.

When we stop to consider that a country such as Norway began to replant and reclaim their forests before Columbus discovered America, it strikes me that it should be a lesson for everyone in this country. Consider too, if you please, that before the war Germany paid her entire road taxes from nothing but the production of nut trees along the public roads. We also know, although a very small country in area, that it produced enough timber each year to satisfy the need for building and commercial purposes in the form of packing cases, casks, etc. And here we are, a country forty times larger than Germany, and forced to depend on countries such as Canada and Norway for wood pulp out of which we manufacture a great many grades of paper.

Some twenty years ago I had a political friend introduce a bill during a meeting of the state legislature, which made it mandatory for the road overseer to plant nut trees along the right of way all over the state; but like many meritorious bills, it was pigeon-holed until the next meeting of the legislature. It seemed an impossibility to resurrect this and an exceptionally fine forestry bill.

Unfortunately I promised to preside at a meeting of conservationists and it is for that reason that I am unable to meet and be with your honorable body, for I would like so much to be permitted in a humble capacity to assist in carrying on the work which you gentlemen are doing, as it is going to mean so much to future generations. I am sure that each of you feels as I do in this matter and that is that "He who serves others, best serves himself."

When the matter comes up for consideration I would like very much to have your next convention here in the Middle West, either in St. Louis or

Alton, Ill., which is only a few miles north of St. Louis and in the vicinity of a splendid nut-producing section, particularly the pecan.

THE CHESTNUT

C. A. Reed, U. S. Department of Agriculture

No discussion of the nut industry in the North at this time would be complete without a brief review of the chestnut situation. The destruction wrought by blight in wiping out practically all of the native chestnut trees within its path, with almost equally fatal results to the European species has for the time being all but eliminated the chestnut from the consideration of planters in the eastern part of the country.

The chestnut bark disease has cost the country untold millions of dollars, and no wonder the public pauses for a second thought before investing in eastern-grown chestnut trees. Nevertheless, it is not to be supposed that chestnut growing has disappeared from this country for all time. No plague has ever been known to wipe a race completely out of existence, and it is unthinkable that the blight will do so with the genus *Castanea*.

The native range of the American sweet chestnut centers largely in the Appalachian region from Portland, Maine, south to Atlanta, Georgia. The species becomes more sparsely represented as the distance increases in any direction from this central area, practically disappearing on the west; in the region of the Mississippi above Memphis. Its northern boundary might roughly be described as extending from lower Illinois through northern

Indiana, southwestern Michigan, southern Ontario, central New York and middle New England. As was to have been expected, the blight has wrought its greatest destruction in places of densest representation of the chestnut species. It is in the outlying districts of scant frequency that the danger of infection from chestnut trees from the forest is least to planted trees, and likewise, there it is that combative measures should be most successful. Obviously, the farther from the center of the native range trees can be planted, the less is the likelihood of infection.

Well outside the native range of the chestnut species, there are a number of districts in the United States within which it should be possible to build up a new chestnut-orchard industry. In proof of this, there are many profitable trees and small orchards in the mid-west and on the Pacific Coast, particularly in western Michigan, northern Indiana, southwestern Illinois, in the eastern foot-hill region of northern California and in the Willamette Valley of Oregon. Probably the most outstanding instance of successful chestnut orcharding now existing in the entire country is a planting of Mr. E. A. Riehl, of Godfrey, Illinois, situated on the bluff of the Mississippi River eight miles west of Alton. Here Mr. Riehl has produced half a dozen or more hybrid varieties which are paying very satisfactory dividends on fertile hillside land which is mainly too steep for cultivation. A number of these varieties have been taken to northern California where they are proving highly successful.

In the Willamette Valley of Oregon, two species are represented with about equal frequency. These are the native chestnut from the eastern states and that from Japan. Neither has performed in such a way as to be particularly encouraging. The former has not been productive and the latter has produced nuts of quality so inferior as to prejudice the planters against the entire genus. It is a difficult matter, therefore, to induce prospective planters in that section to consider any species of chestnut.

In the East, it is well known that the native species does not come into bearing until 12 or 15 years of age at best, and that to induce pollination and a set of nuts, it is necessary to inter-plant a number of varieties together. Had groups of varieties of American or European origin been planted on the Coast, instead of single trees of the former or varieties from Asia, it is not improbable that the present attitude toward the chestnut in the Pacific Northwest would have been quite different.

The work of the late Dr. Van Fleet, in hybridizing various chestnut species and in testing out Chinese and Japanese species with a view to determining their value as nut producers and their resistance to the bark disease, is familiar to most members of the Northern Nut Growers' Association. Since the death of Dr. Van Fleet, the work has been taken over by other hands in the Bureau of Plant Industry; but apparently, all of the hybrids now growing in the vicinity of Washington, D. C., are destined to succumb to blight. At present, practically every tree of the Chinese chestnut *Castanea molissima*, planted by Dr. Van Fleet at Bell Station, Maryland, where his work was mainly centered, likewise shows large blight cankers. But despite the gravity of the infections, it does not appear wholly improbable that many of these trees can be preserved. However, the wisdom of continuing propagation of the Japanese species is very doubtful, as the quality of nuts is usually of low order. Chestnut trees from China are generally light producers; but out of the total of several hundred at Bell, several this year have borne good crops. The flavor of the nuts is sometimes sweet, but oftener, otherwise; yet the average is superior to that of the Japanese chestnuts produced in the same orchard. Fortunately, it happens that the nuts from some of the trees of Chinese species which have been most prolific during the past season, have proved to be of high quality, comparing favorably in this respect with the native sweet chestnut. In size, the Chinese chestnuts average much above those of the American species, and while perhaps a shade smaller than those from Europe, they are of a size and quality which should readily appeal to market demands.

An early planting of Chinese chestnut trees at Lancaster, Pa., put out by Mr. J. F. Jones, Vice-President of the Northern Nut Growers' Association, proved so susceptible to blight that all were subsequently destroyed. On the other hand, not infrequent reports are reaching the Federal Department of Agriculture of instances in which the species is shown to be highly resistant, even when grown within blight-affected districts. Secretary Deming is one of those from whom reports of this kind have been received. His planting, consisting of 12 trees put out in 1915 near Georgetown, Conn., has recently borne some nuts. Other cases, some reporting one way and others the other, might be cited; but let it suffice to say that the chestnut industry, although temporarily set back seriously, is not necessarily doomed.

REPORT OF THE COMMITTEE ON NOMENCLATURE

C. A. Reed, Chairman

While no new names of varieties appear to need consideration at this time, it may be well for the Association to refresh its memory regarding a few of the outstanding rules of the standard code of nomenclature by which the Society is guided in the recognition of names. In common with practically all other leading horticultural organizations of the country, including the National Pecan Growers' Association of the South, the Northern Nut Growers' Association follows the code of nomenclature of the American Pomological Society. Some of the provisions of this code are substantially as follows:

> 1. A name shall consist, preferably, of but one word, although under specified circumstances, two words may be permitted.
>
> 2. In selecting a name, "The paramount right of the originator, discoverer or introducer of a new variety within the limitations of this code, is recognized and established."
>
> 3. A name shall be recognized as fixed and shall have the right of priority over any others subsequently applied, after having appeared

in print in such a way as to be definitely tied to a variety, or established.

These references call attention to the fact that the code does not define the meaning of the term "variety," and as it does not appear that a clear cut definition has appeared elsewhere in recent literature, in modern application, it may be well to state how it is being interpreted by this committee.

In horticultural practice a plant is not regarded as acquiring varietal status until it becomes distinctive among seedlings, because of superiority of product, unusual history, or other similar reason. Few tree varieties are recognized as such until after having been propagated by at least one asexual method, such as budding, grafting, layering or dividing.

The Committee calls special attention to a recent report on nomenclature, appearing in a bound volume of 546 pages, under the title "Standardized Plant Names." This report was prepared and published by the American Joint Committee on Nomenclature, which was duly appointed by the leading horticultural societies of the country. It represents the latest authority on matters of horticultural nomenclature, and is indorsed by the leading horticultural authorities of the present time. Of immediate interest to this Association is the fact that *Hicoria* replaces *Carya* as being the proper generic name of the hickory group.

NOTES FROM AN EXPERIMENTAL NUT ORCHARD

Willard G. Bixby, Baldwin, N. Y.

For several years the association has been advocating the planting of experimental nut orchards, and ever since I heard of this suggestion I have been desirous of having one and being able to contribute information to our knowledge of nut growing. Therefore since 1917 I have been assembling at Baldwin material which I hoped would aid in this. At the Rochester meeting some of the results were noted, and this year, I trust, something presented will prove of interest.

CHESTNUTS—Last year I expressed the belief that by carefully watching chestnut trees and cutting out the blight as soon as it appeared it should be possible to grow and fruit almost any variety in the blight area. This I have done with every variety that I have, but that is about all, apparently, that it is possible to do, for nearly all of my trees have been badly attacked by the blight at the crown; that is at the junction of the root and trunk, and to cut out the blight means to cut down the tree. The most resistant variety noticed so far is the Boone, which has some Japanese chestnut parentage, but probably the Boone trees will not last over a year longer.

Apparently it is going to be necessary to get some resistant stock and do the grafting high enough to prevent fatal attack of the blight at the crown. Mr. P. W. Wang sent some Chinese chestnuts in the fall of 1921, and I have now several hundred seedlings of what I suppose are Castanea mollissima, of which I plan to grow a number to rather large size, set them out where the next planting of chestnut trees is to stand, and graft the branches to fine varieties. It will take at least two or three years, however, before this can be done.

HAZELS—For some four years I have been assembling, for hybridizing purposes, selected American hazels from various sections of the United States as well as the various European cultivated varieties that gave promise of being hardy. This year both blossomed rather freely, but the only variety of which I had enough pollen to work with was the Italian Red. The staminate flowers were picked from some six or eight American hazels which were blooming well and the pistillate flowers were pollinated with Italian Red pollen, in the hope that some hybrid nuts would result. Although the pollination was repeated twice I was much disappointed to find only an occasional nut as a result.

It is to be said in this connection, however, that there were practically no nuts on these American hazels which had not been pollinated with strange pollen; so the lack of nuts could not be laid to the artificial treatment given the flowers of those plants where it had been planned to make hybrids. Apparently it was due to climatic conditions that nuts were almost lacking on all hazels here this year; but I do not recall any severe cold spells when the hazels were in flower. Still, on one or two branches which I had tagged, as being particularly full of pistillate flowers, there were noticed an almost equal number of dead pistillate flowers a little later. It is seemingly going to be well to carefully study the development of the hazel flowers into nuts. They grow differently from the walnuts and the hickories. The hazel flowers apparently, after being fertilized, develop into stems on which the existence of nuts escapes the attention, at least of the casual observer, until about August, while the nuts on the walnuts and the hickories even though small at first, are plainly visible from the time they are formed by fertilized flowers until they are matured.

HICKORIES—The bearing age of the transplanted hickory so far has been almost an unknown quantity, and what we did know has been such that the association has hesitated to say much about planting hickories, its recommendations on the hickory being confined to that of topworking existing hickories. These are known to begin bearing soon after topworking, records of bearing in two or three years not being unusual.

On transplanted hickories, however, about all the information of which I know is as follows: The late Mr. J. W. Kerr, of Denton, Md., many years ago bought a number of shagbark hickories from a nursery, set them out and noted that the time that elapsed before they bore was about 25 years. Mr. Rush's Weiker tree, which bore in 11 years after being set out, cut down this time materially.

A Kentucky hickory on my place set out in the fall of 1917, flowered this year, but I had no pollen with which to fertilize the blossoms, and the nutlets dropped off. A young shagbark seedling set in its present location in the fall of 1919 and grafted to Barnes this spring, also set a nut, but this dropped off like those on the Kentucky and apparently for the same reason. It would certainly seem as if under favorable conditions, the transplanted hickory is not going to be anywhere near as slow as feared in coming into bearing.

WALNUTS—A Royal and a Paradox walnut each supposed to be grafted trees with scions from Burbank's original trees, bloomed this year, and the Royal has a number of nuts on it. The Paradox has been here a very much shorter time, not over two or three years; so perhaps it is too soon to be expecting nuts. The Paradox is said to be a very shy bearer, setting nuts only occasionally, and then but few; still, one of my Paradox trees which is not over three feet high, blossomed full. It would seem as if it might pay to study this tree and see if the sterility or fancied sterility of this tree could not be overcome by seeing that proper pollen is at hand at the right time. A Cording walnut, a hybrid between the English walnut and the Japan walnut not quite 3 feet high, is bearing a nut this year.

Grafting—Perhaps the most interesting thing to be related is the result of attempts to determine the species of hickories best suited as stock for the fine varieties of hickories that we have. In preparation for this and through the kindness of Mr. Henry Hicks of Westbury, L. I., over 100 each of hickory trees of several species were obtained and set out in the fall of 1919. They were in fine condition for grafting this spring. There are some fifteen species of hickories native in the United States. The fine varieties of hickories that we have which are generally supposed to be largely shagbarks may prove to be much better adapted for grafting on some stocks than on others. A knowledge of this will prove to be of great value in top working. The grafting was done by Dr. Deming, on May 29, 30,31 and June 1 of this year, 31 grafts being set on shagbark stock, 52 on mockernut, 53 on pignut, 47 on pecan and 91 on bitternut, a total of 274. There were also 343 walnut grafts set on walnuts of four species. The results of this work are summarized in the tables following:

HICKORIES, SCIONS FROM YOUNG TREES

Stocks, Number of Grafts and Per Cent of Catches

	Bitternut		Mockernut[1]		Pecan		Pignut		Shagbark		Total	
Variety	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Barnes, scions												

Dr. Deming's trees 3 100.0 3 100.0 3 100.0 3 100.0 6 100.0 18 100.0
Gobble, scions
Dr. Deming's trees 1 100.0 1 100.0 1 100.0 1 100.0 1 0.0 5 80.0
Griffin, scions
Dr. Deming's trees 1 100.0 1 0.0 1 0.0 1 100.0 1 100.0 5 60.0
Hales, scions
W. G. Bixby's trees 5 100.0 5 60.0 5 80.0 4 25.0 19 68.4
Kirtland, scions
Dr. Deming's trees 3 66.7 3 33.3 3 66.7 3 66.7 12 58.3
Laney, scions
Dr. Deming's trees 6 66.7 6 66.7
Long Beach, scions
Parent Tree 3 33.3 3 66.7 4 50.0 4 25.0 3 100.0 17 53.0
Siers, scions
Dr. Deming's trees 5 100.0 5 100.0
Stanley, scions
Dr. Deming's trees 3 66.7 3 66.7 3 66.7 9 66.7
Taylor, scions
Dr. Deming's trees 4 75.0 5 60.0 5 80.0 3 100.0 17 86.5

Total 34 80.8 24 60.8 22 68.1 22 72.9 11 75.0 113 74.0

[Footnote 1: The mockernuts were larger than any other hickories grafted excepting some bitternuts referred to in the next footnote. They were grafted mostly on branches.]

HICKORIES, SCIONS FROM OLD TREES

Stocks, Number of Grafts and Per Cent of Catches

Bitternut Mockernut[2] Pecan Pignut Shagbark
Total

Variety No. % No. % No. % No. % No. % No. %
Brooks, scions from parent tree,
  poor condition 5 40.0 5 0.0 5 20.0 5 40.0 20 20.0
Clark, scions from parent tree,
  poor condition 5 40.0 5 0.0 5 20.0 5 40.0 5 20.0 25 20.0
[3]Fairbanks, scions from
  parent tree (?), dry but
  otherwise good 27 57.8 27 57.8
Kentucky, from parent tree,
  poor condition 5 20.0 3 33.3 5 80.0 5 80.0 5 80.0 23 60.8
Manahan, scions from parent
  tree, poor condition 5 20.0 5 0.0 5 20.0 6 33.3 5 20.0 26 24.6
Vest, scions from parent tree,
  poor condition 5 20.0 5 0.0 5 40.0 5 60.0 5 20.0 25 20.8
Weiker, scions from parent
  tree 5 20.0 5 0.0 5 60.0 15 26.8
                                 —— ——— —— ——- —— ——— —— ——— —— ——— ——- ———
Total 57 45.0 28 5.5 25 36.0 31 45.6 20 35.0 161 32.9

[Footnote 2: The mockernuts were larger than any other hickories grafted excepting some bitternuts referred to in the next footnote. They were grafted mostly on branches.]

[Footnote 3: Of these scions 5 were set in branches on two trees 1-1/4 or so in diameter and showed 100% catches; balance were set in the top on small trees 1/2 diameter or less, and showed 54.5% catches.]

BLACK WALNUTS, JAPAN WALNUTS, PERSIAN WALNUTS BUTTERNUTS

Stocks, Number of Grafts and Per Cent of Catches

Black Walnut Butternut Japan Walnut Persian Walnut
Variety No. % No. % No. % No. %

Adams Black Walnut, scions
parent tree 13 15.4
Alley Black Walnut, scions
parent tree 9 0.0
O'Connor Hybrid Walnut, Persian
Walnut and Black Walnut (?)
scions parent tree 9 22.2
—- ——
31 12.9

Ohio Black Walnut, scions W. G.
Bixby's trees 17 64.7
McCoy Black Walnut, scions W. G.
Bixby's trees 9 77.0
Stabler Black Walnut, scions some
W. G. Bixby's trees, and some Dr.
Deming's trees 85 51.2
[4]Ten Eyck Black Walnut, scions
W. G. Bixby's trees 32 97.0
Thomas Black Walnut, scions W.
G. Bixby's trees 23 100.0
Wasson Black Walnut, scions W.
G. Bixby's trees 8 75.0
—- ——
174 69.5

Persian Walnuts 4 varieties, scions
about 2-3 from parent trees, all

of which were quite vigorous
growers 46 0.0
Aiken Butternut, scions W. G.
Bixby's trees 39 38.5
Lancaster Heartnut, scions W. G.
Bixby's trees 53 3.8

[Footnote 4: One scion was overlooked in tying and waxing, otherwise apparently we would have had 100% catches.]

* * * * *

In the above two groups of hickories the one where scions were cut from young, rapidly growing trees, contrasts unmistakably with those where scions were cut from old bearing trees. The same is shown in the table of black walnut grafts, where the Alley, Adams, and O'Connor scions were cut from old bearing trees, and the others from young, rapidly growing trees.

The poor success with the heartnuts is quite in line with previous attempts at propagating this species by grafting. Results shown here with the butternut are deemed reasonably satisfactory, in view of the well known difficulty of grafting this species. It should be noted here that, in the case of every graft that took and grew, it was the small buds that were successful, not the large ones. The total lack of success with the Persian walnut is inexplicable to the writer, but he knows of no previous attempts to graft Persian walnut on Persian walnut root.

Black walnuts show a very high percentage of catches, in the case of the Thomas and Ten Eyck varieties 100%, but in the case of the Stabler this is reduced to 51.2%. I would say in this connection that neither of my two Stabler trees are vigorous growers, and so the trees grafted with scions from these are really cases where we have not been using scions from vigorous growing trees, and we know that this does not give a high percentage of catches.

The proper species to be used as a stock for the various varieties of hickories has not been shown conclusively for the number of grafts of each kind set was too few to be conclusive, and these experiments should be repeated. In the case of most of these varieties where results are poor, it was particularly noted when the grafts were set that the scions were in poor condition, a number of scions being thrown away because the cambium layer was dead. It is to be hoped that a species will be found to which will be well adapted the Vest hickory, which the writer regards, everything considered, as the best hickory that we have. Seemingly the pecan is the stock that gets the greatest number of catches; but the difficulty the writer has had in making Vest hickories on pecan root live, leads him to question as to whether another stock might not prove better. Another thing disappointing so far is in the seeming poorness of the mockernut as a stock. Over quite a large section of the United States the mockernut is the prevailing hickory, and in that section the mockernut will be most generally available for top working; moreover it will grow well in sandy soils where the shagbark is not found. In Petersburg, Va., the writer has seen it seemingly outgrow the black walnut.

The adaptability of the Barnes hickory on all stocks is notable, for it is the only one of the 10 fine hickories tested in the 1919 contest, of which this is true. If these grafts continue to flourish, and especially if future experiments check the results this year, the Barnes will have a peculiar value for top working. It is one of our best hickories, and, apparently is our surest variety for top working.

MR. CLOSE: I would suggest that we extend our thanks to the Smithsonian

Institute for the use of this room for the meeting.

THE PRESIDENT: Will you vote for that? (Motion voted upon favorably). I believe then, that brings to a close the Fourteenth Annual Convention, to meet in New York for the Fifteenth Convention in 1924, on September 3,4 and 5.

This meeting is now adjourned.

Time—2:30 p. m.

* * * * *

Notes of this convention by Mrs. B. W. Gahn, U. S. Dept. of Agr., Washington, D. C.

APPENDIX

Among those present were the following:

Senator Penney—Saginaw, Michigan.
B. K. Ogden—3306 19th St., N. W., Washington, D. C.
W. G. Slappey—12 Boyd Avenue, Takoma Park, D. C.
S. von Ammon—Bureau of Standards, Washington, D. C.
A. M. Greene—Ridge Road, N. W., Washington, D. C.
Alfred Heine—Bowie, Md.
H. Harold Hume—Glen St. Mary, Fla.
R. H. Hartshorn,—Washington, D. C.
Wm. S. Linton—Saginaw, Mich.
W. E. Safford—Botanist, Bureau of Plant Industry, Washington, D. C.
Dr. M. B. Waite—Federal Insecticide and Fungicide Board, Bureau of Plant Industry, Washington, D. C.
Dr. Oswald Schreiner—Bureau of Plant Industry, Washington, D. C.
Karl Wallace Greene—Washington, D. C
C. A. Reed—Bureau of Plant Industry, Washington, D. C.
Mrs. C. A. Reed—Washington, D. C.
C. P. Close—Bureau of Plant Industry, Washington, D. C.
Mrs. C. P. Close—Washington, D. C.
W. R. Mattoon—Forest Service, Washington, D. C.
Thomas P. Littlepage—Washington, D. C.

John M. Littlepage—Washington, D. C.
Eunice M. Obenschain—Hotel Monmouth, Washington, D. C.
J. M. Richardson—Stormville, N. Y.
Robert T. Morris—114 E. 54th St., N. Y.
Dr. Llewellyn Jordan—100 Baltimore Ave., Takoma Park, Md.
Alfred V. Wall—2305 W. Lanvale St., Baltimore, Md.
Jacob E. Brown—-Elmer, N. J.
Albert R. Williams—Washington, D. C.
Mrs. B. W. Gahn—Dept. of Agriculture, Washington, D. C.
James S. McGlennon—Rochester, N. Y.
Ralph T. Olcott—Rochester, N. Y.
Zenas H. Ellis—Fair Haven, Vt.
G. A. Zimmerman, M. D.—Piketown, Pa.
G. F. Gravatt—Forest Pathology, Bureau of Plant Industry, Washington, D. C.
Willard B. Bixby—Baldwin, N. Y.
John W. Hershey—Banks, Pa.
P. H. O'Connor—Bowie, Md.
John E. Carmoday—Charlottesville, Va.
Mrs. John Carmoday—Charlottesville, Va.
Mrs. W. N. Hutt—"The Progressive Farmer," Southern Pines, N. C.
Ammon P. Fritz—55 E. Franklin St., Ephrata, Pa.
W. A. Orton—Bureau of Plant Industry, Washington, D. C.
J. C. Corbett—Dept. of Agriculture, Washington, D. C.
W. G. Pollaret—The Star, Washington, D. C.
Prof. Lumsden—Federal Horticultural Board, Washington, D. C.

EXHIBITS LISTED

Crops of 1923

Exhibit of Robt. T. Morris 1. Hybrid chinkapin (burrs and nuts). 2. Graft of pear tree (paraffin method).

Exhibit of C. A. Reed
"Rush" American Hazel.

Exhibit of C. P. Close 1. Seedling filbert. 2. "Van Fleet" hybrid chinkapin. 3. "Glady" walnut.

Exhibit of J. F. Jones Persian Walnuts. 1. Wiltz Mayette. 2. Meylan. 3. Lancaster. 4. Lancaster (Same). 5. Eureka. 6. Hall. Pecans. 1. Posey. 2. Busseron. 3. Niblack. Hazels. 1. Rush (Three exhibits). Cobnut. 1. (No name). Filberts. 1. Fichtendersche. 2. Daviana. 3. Blumenberger. 4. Italian red. 5. Lambert nut. 6. Friehe Longe. 7. Gunzelebenner. 8. White Aveline. 9. Grosse Ronde. 10. Barcellona. 11. Spanik Gr. 12. Prolific. 13. Noce Lunghe. 14. Du Chilly. 15. Grant de Halle. 16. Buttners. Exhibit of W. G. Bixby 1. Lancaster Heartnuts. 2. Royal Walnuts. 3. Hall Persian Walnuts. 4. Rush Persian Walnuts. Exhibit of T. P. Littlepage (Grown on his farm). 1. Chinkapins. 2. "O'Connor" walnuts. 3. Mixture of varieties of European filberts. 4.

Cluster of pecans (Indiana). 5. Littlepage hazels (which Mr. Littlepage called "American"). 6. Spanish chestnut.

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